Suspension stiffness and damping comparison

Each experiment changes just one constant; all other model settings and initial conditions are held fixed.

Fixed motor-to-wheel reduction: 11.5:1. Baseline is rerun with the four experiments.

No retrimming: startup differences include settling from the same initial state and unchanged preload.

Suspension travel is recorded relative height; body Z is positive downward. Body acceleration uses 9.80665 m/s^2 per g.

Curves stop at each run's recorded end. Failure / incomplete runs do not receive valid lap times.

Comparisons use simulation time: different lap pace can shift cornering events between curves.

Extra completed-case views omit the unstable case so smaller differences remain visible.

Baseline: K = 35000 N/m; C = 2500 N*s/m; Completed; end time 34.790 s.

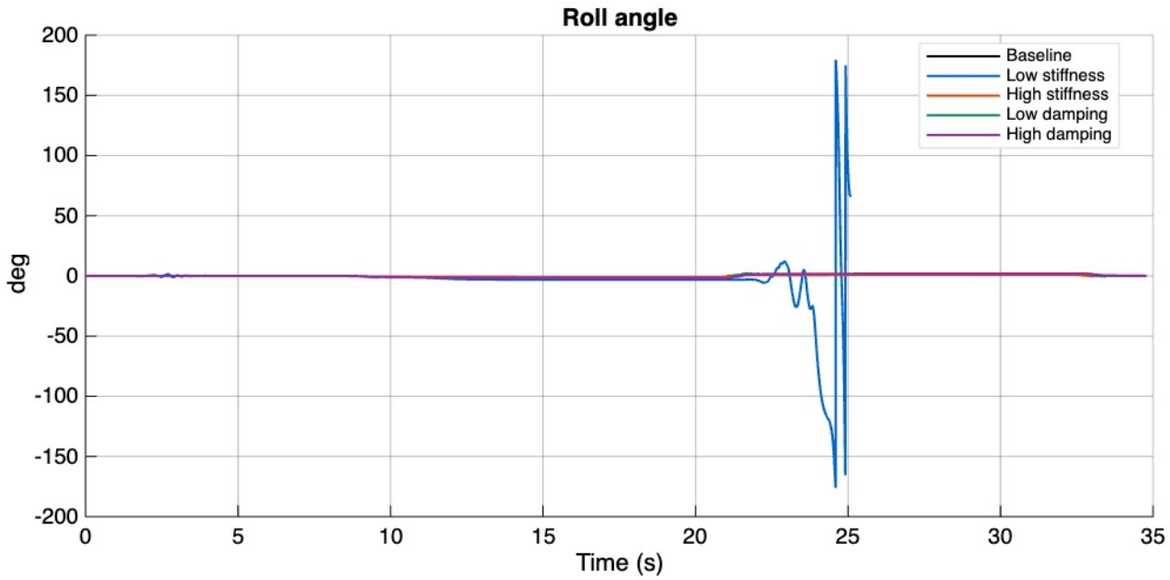
Low stiffness: K = 3500 N/m; C = 2500 N*s/m; Incomplete; end time 25.090 s.

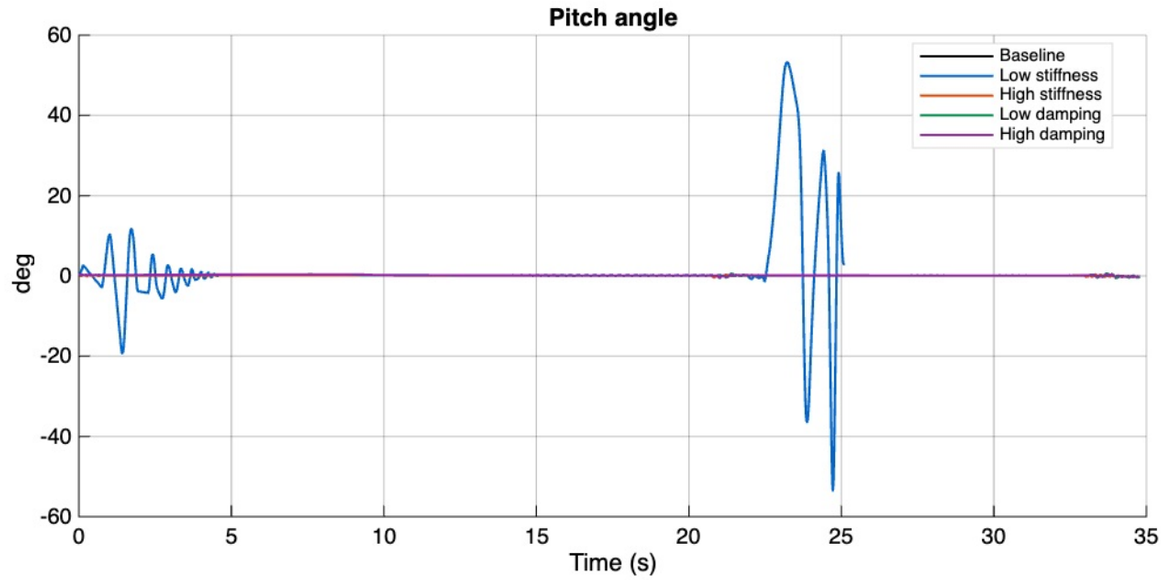
High stiffness: K = 350000 N/m; C = 2500 N*s/m; Completed; end time 34.431 s.

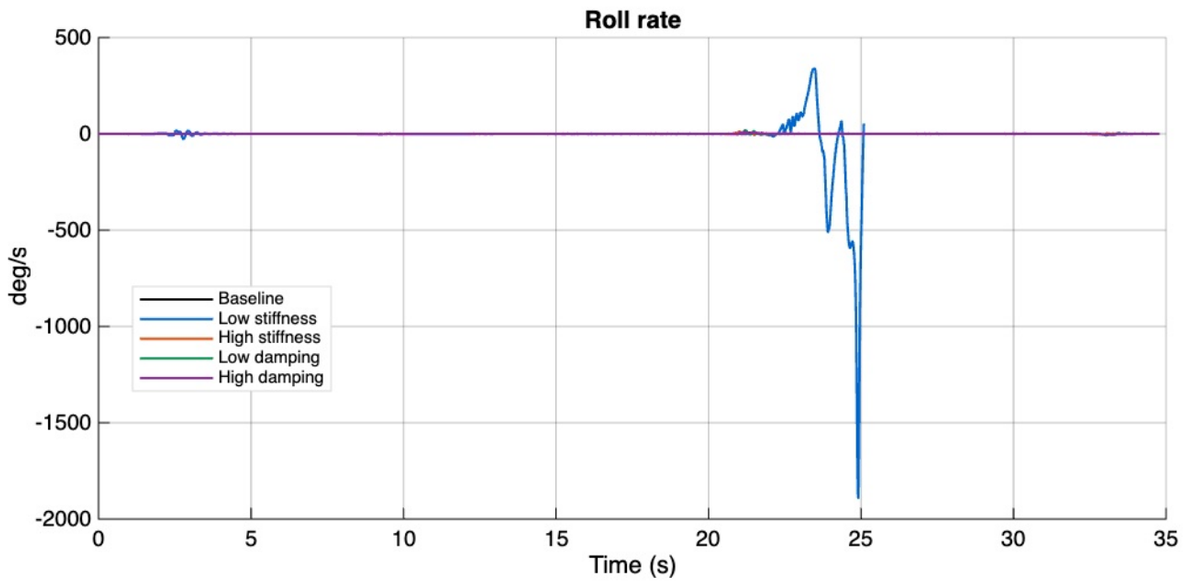
Low damping: K = 35000 N/m; C = 250 N*s/m; Completed; end time 34.792 s.

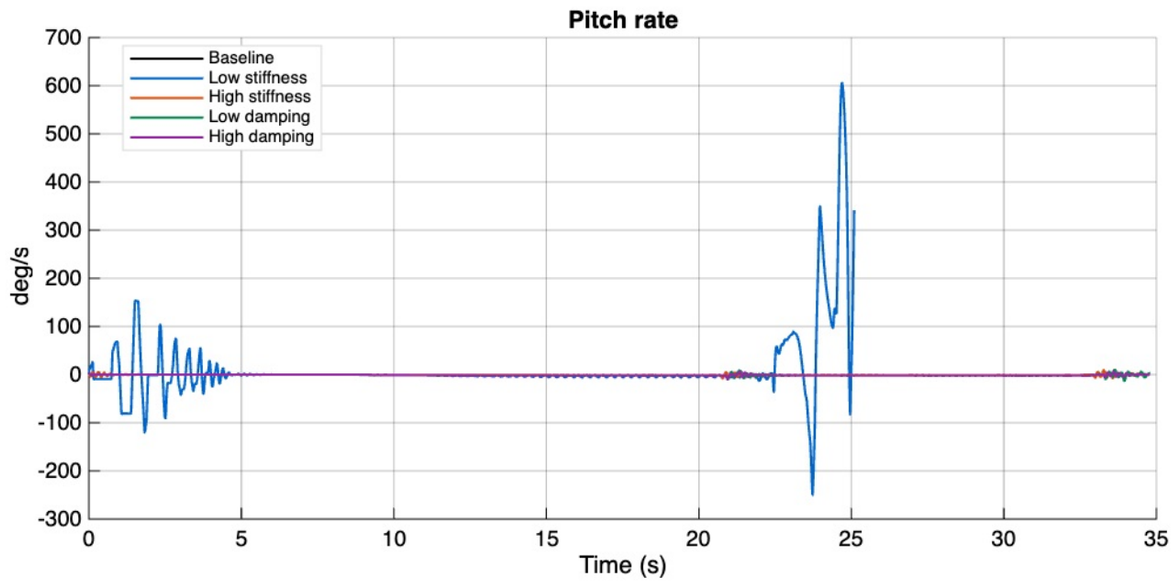
High damping: K = 35000 N/m; C = 25000 N*s/m; Completed; end time 34.787 s.

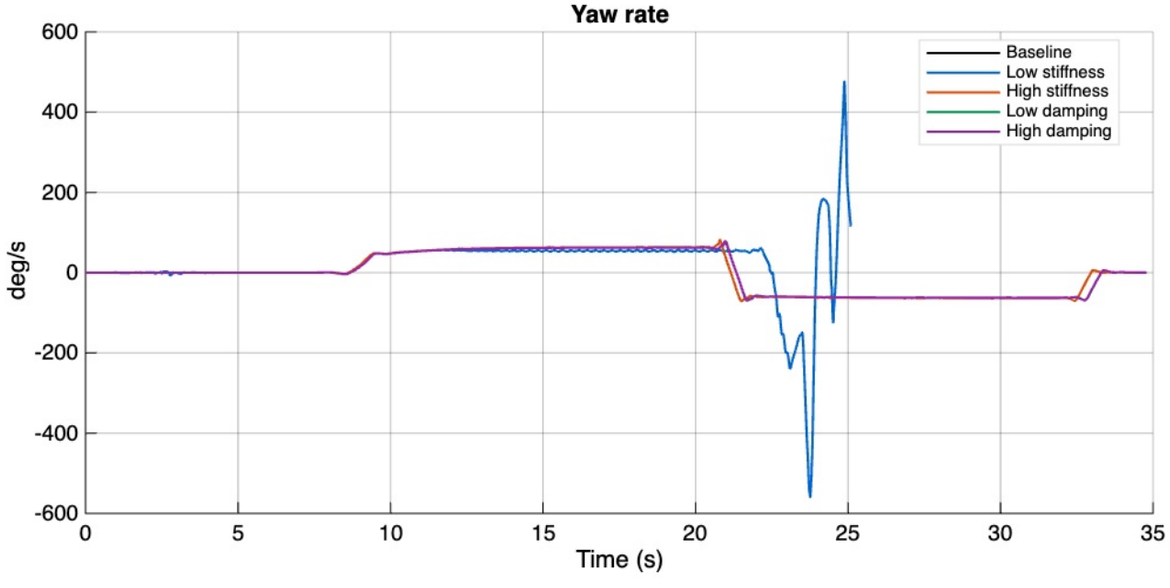
Suspension comparison - page 1

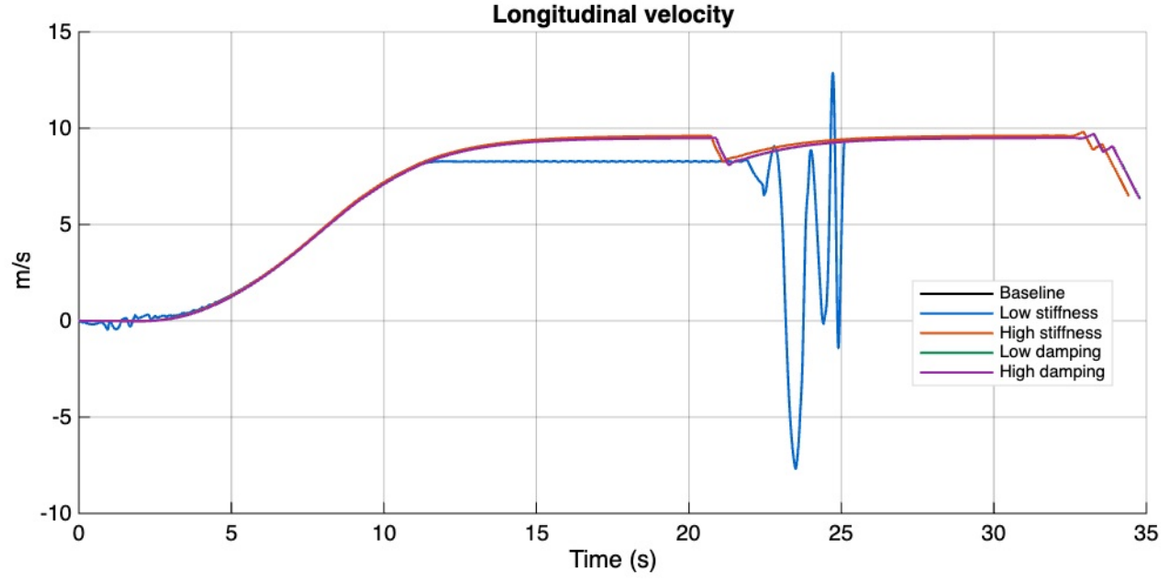




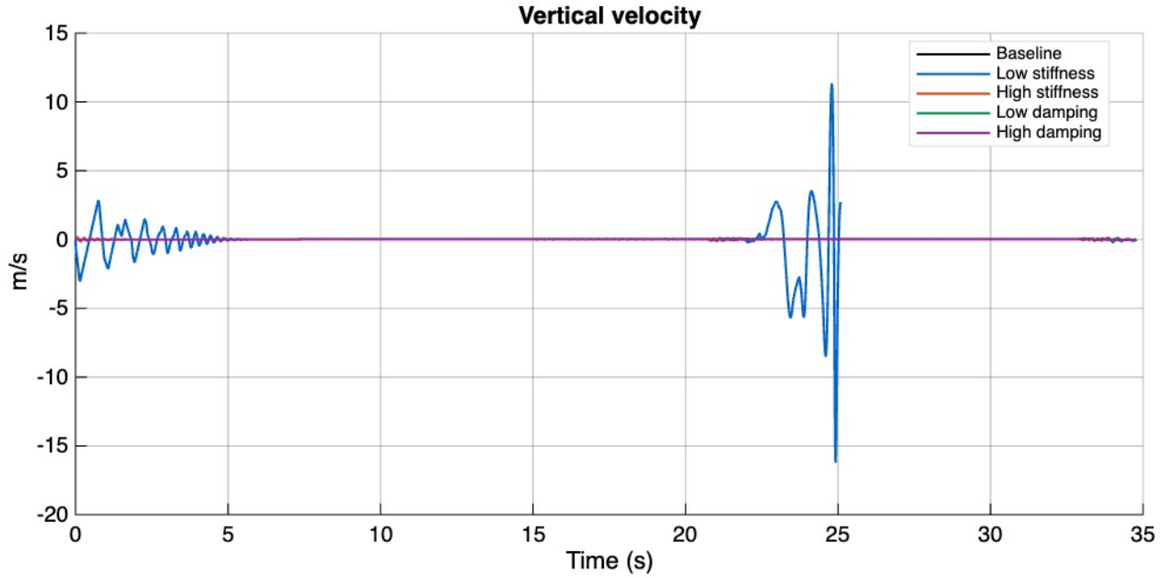


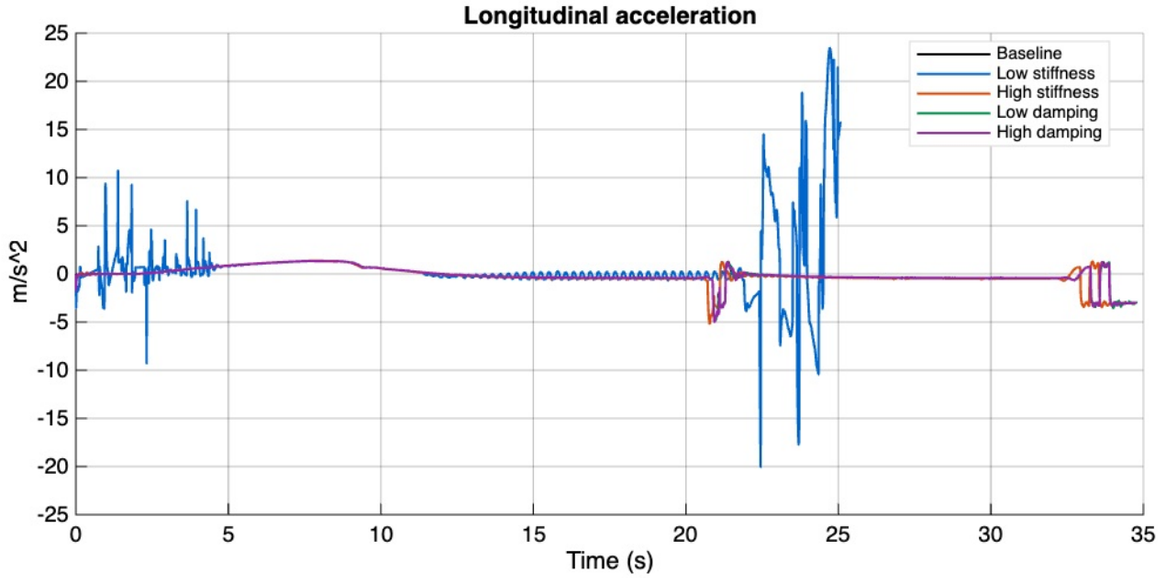


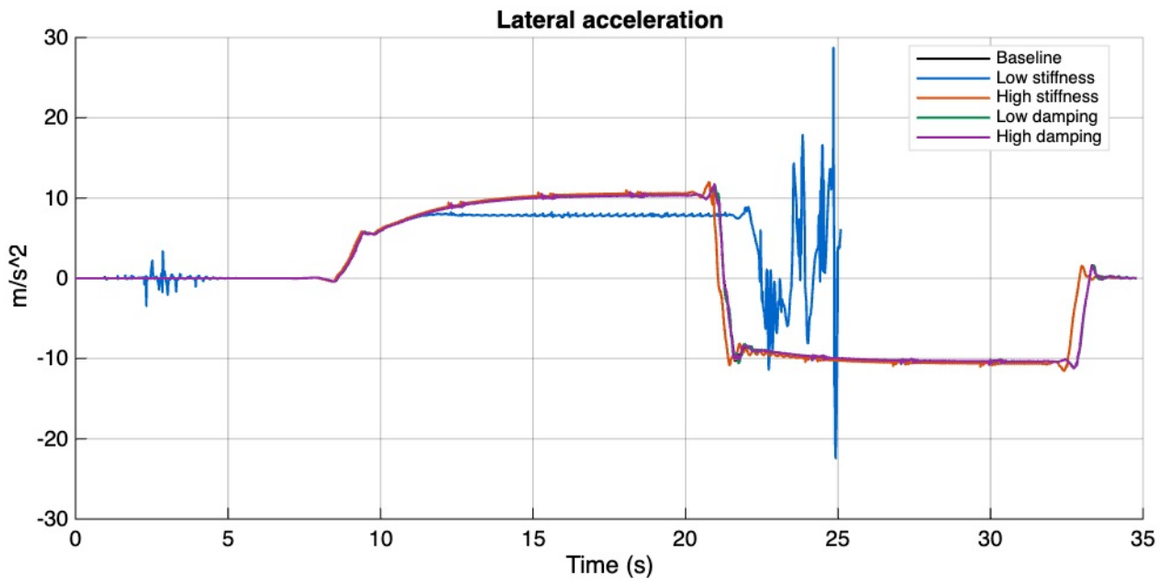


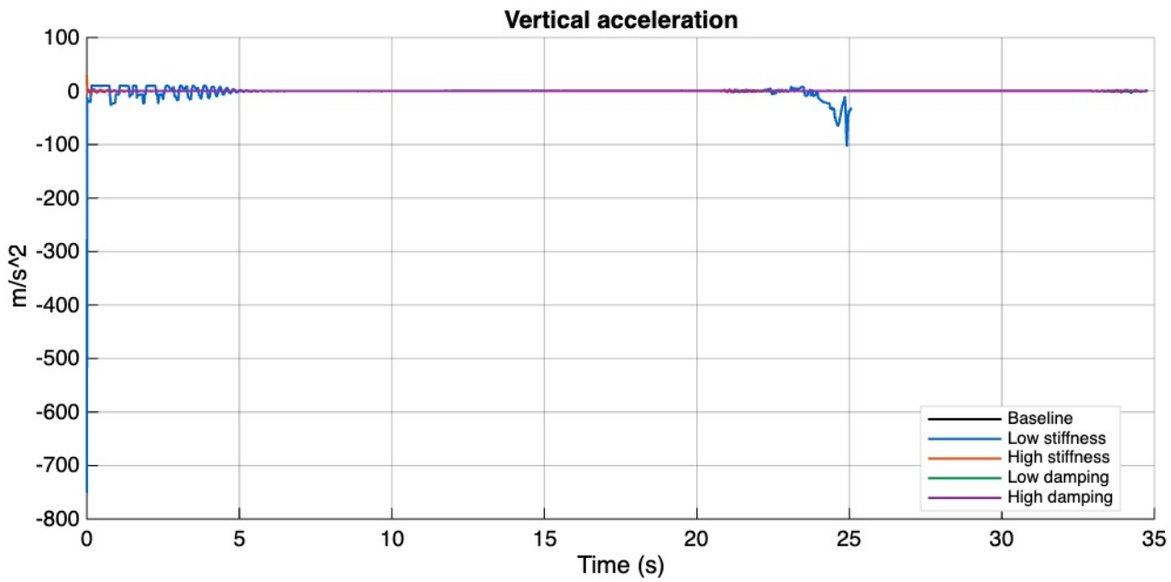


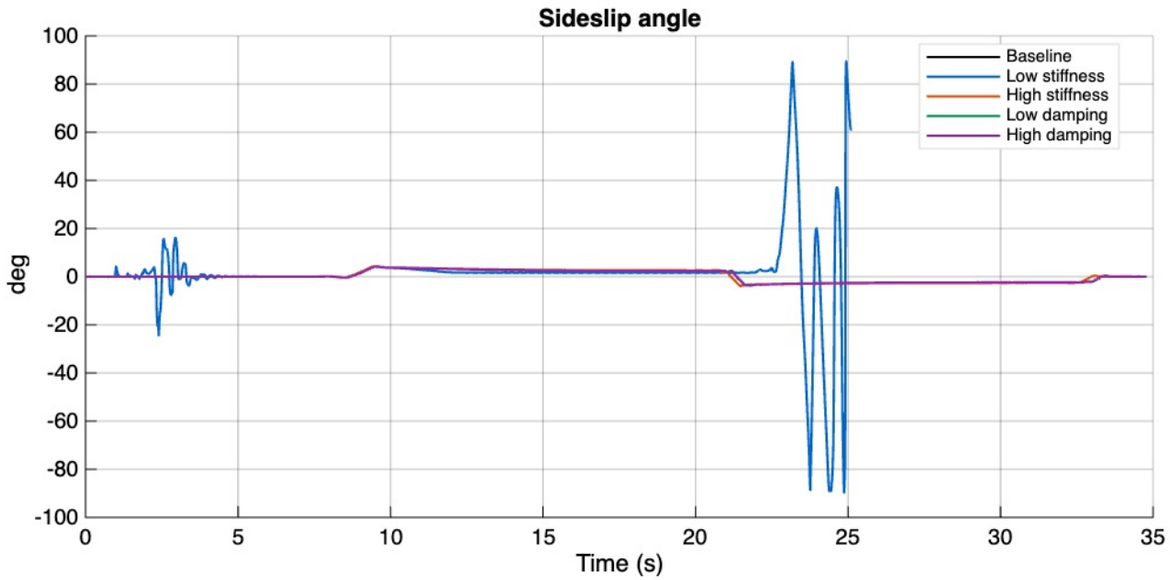
Suspension comparison - page 2

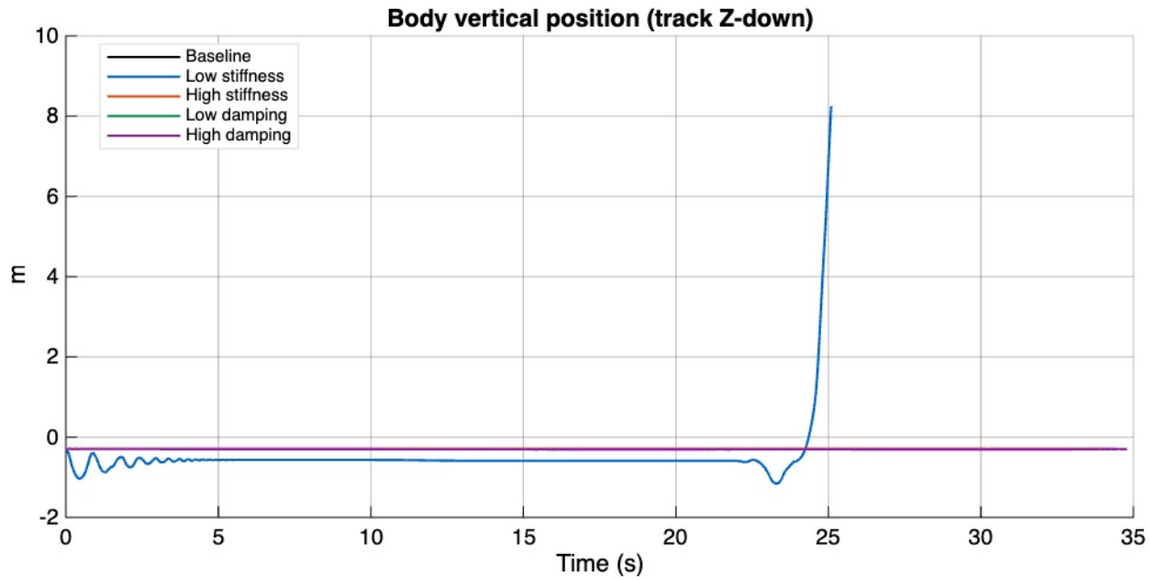




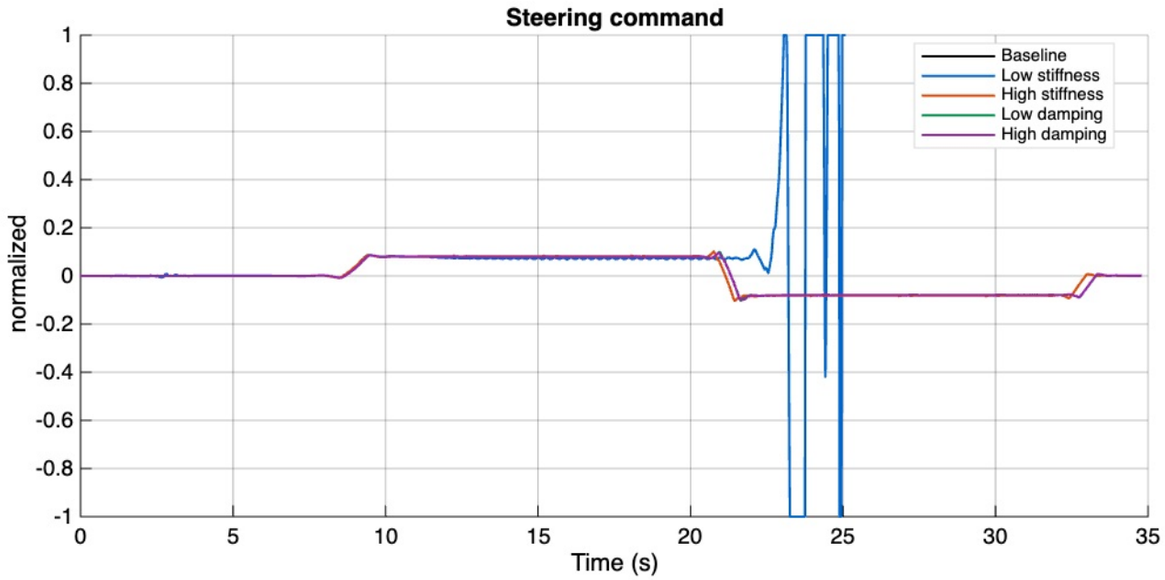


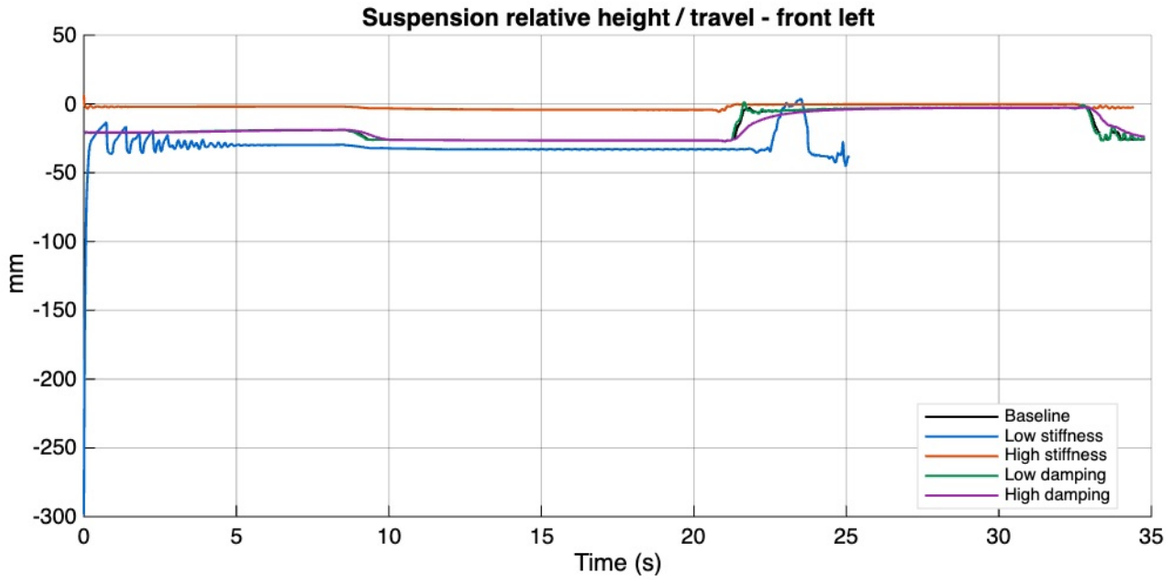


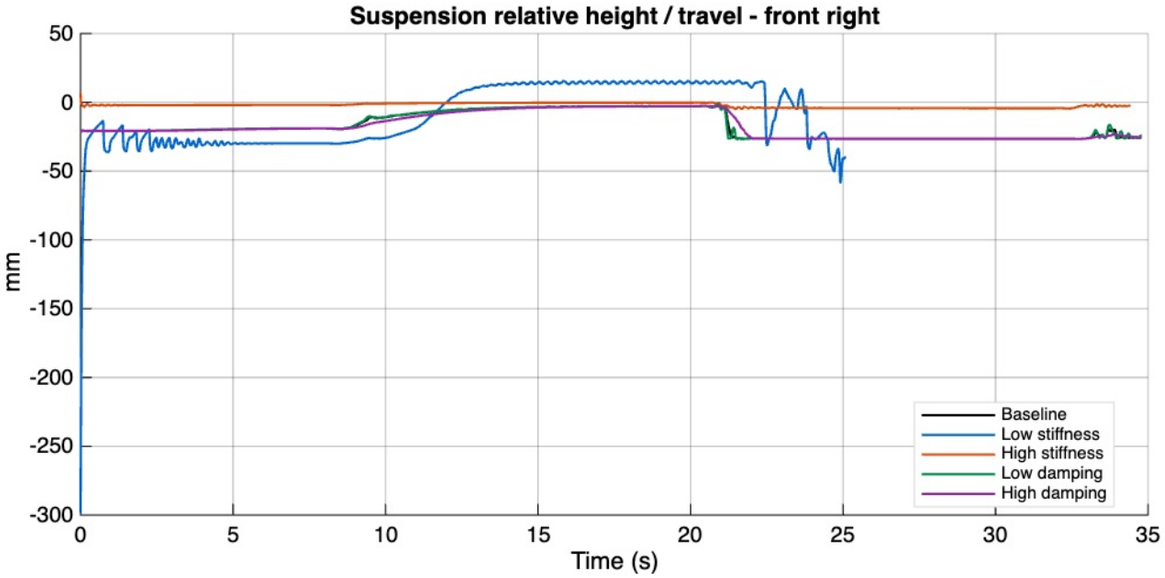


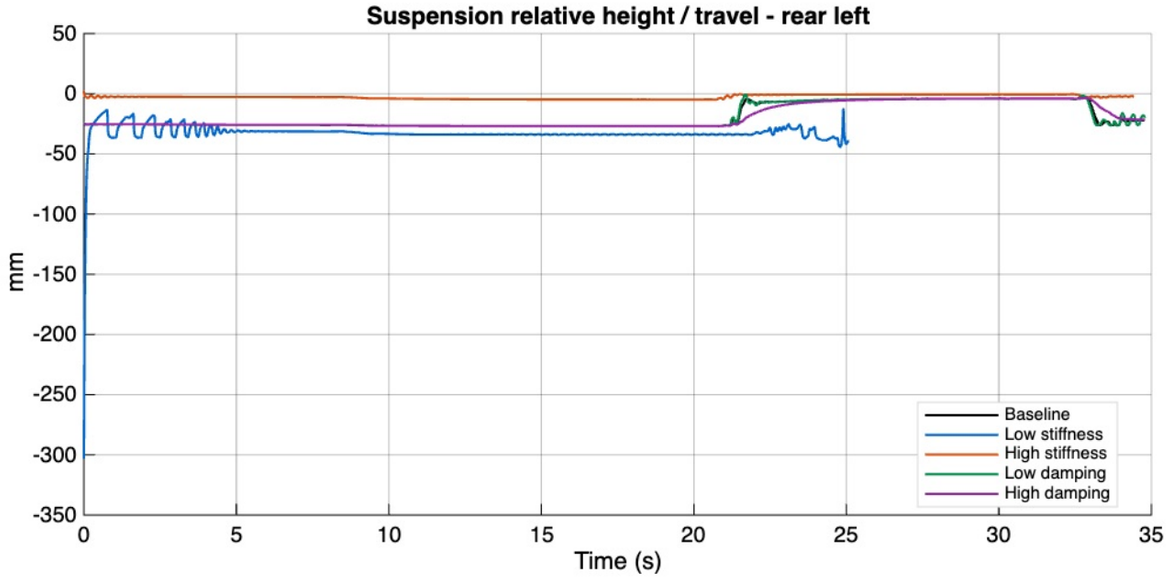


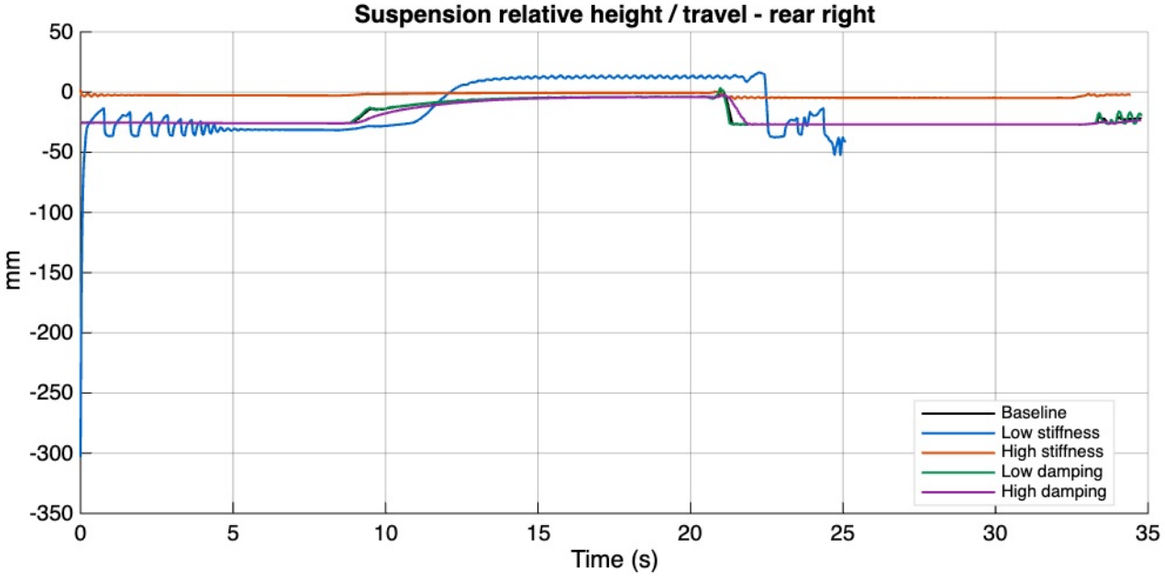
Suspension comparison - page 3

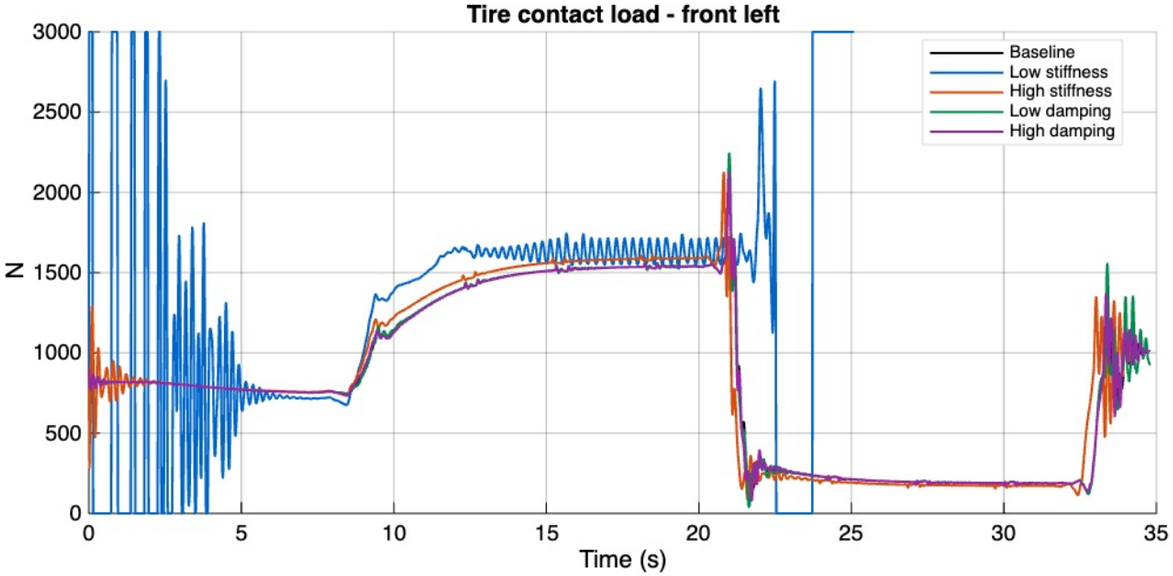




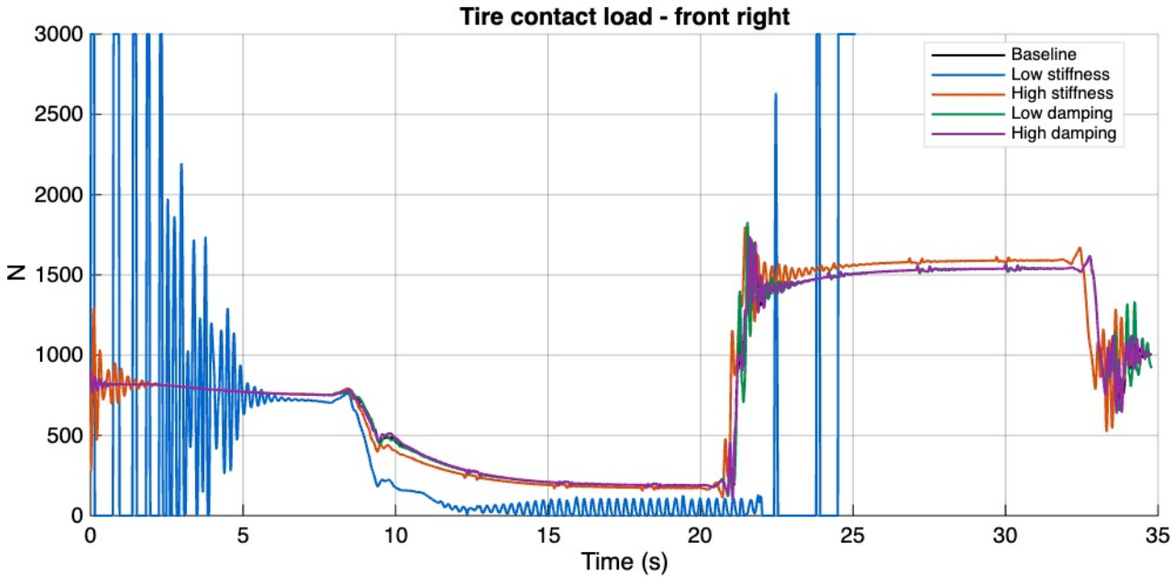


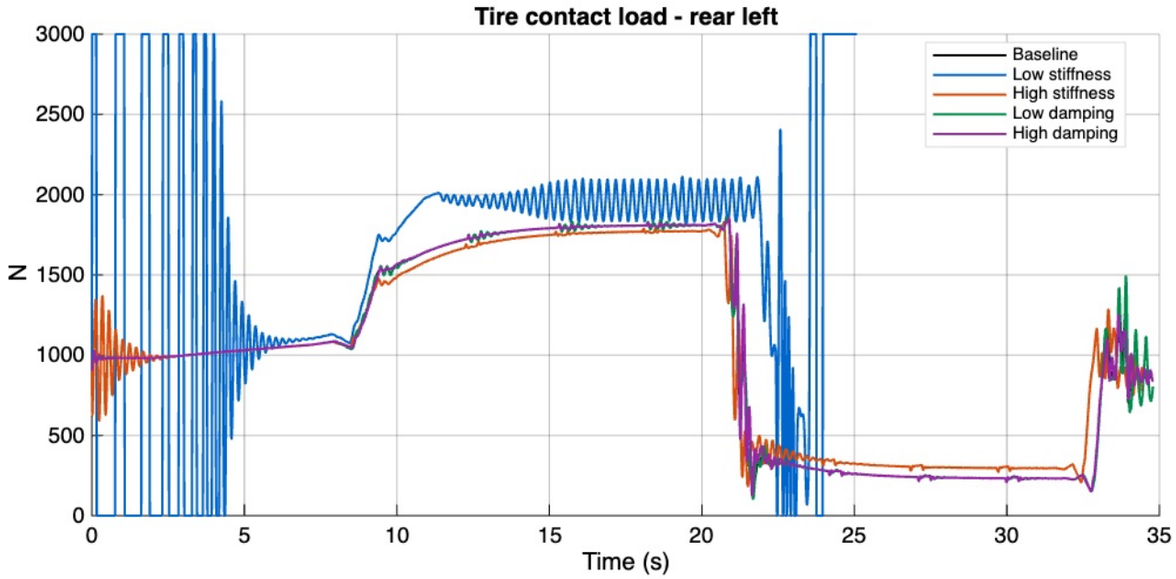


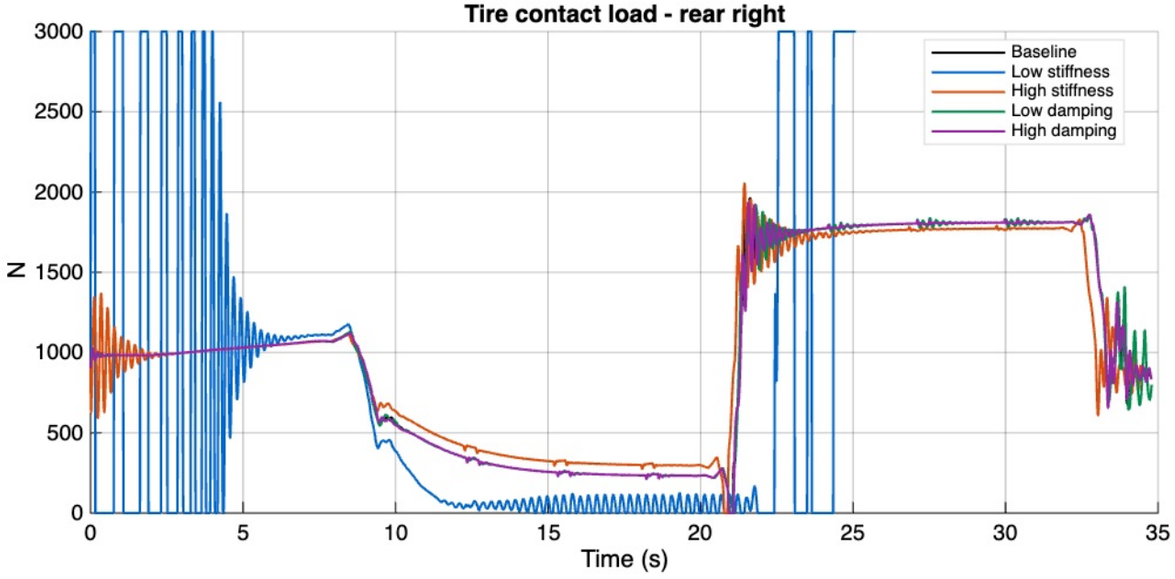


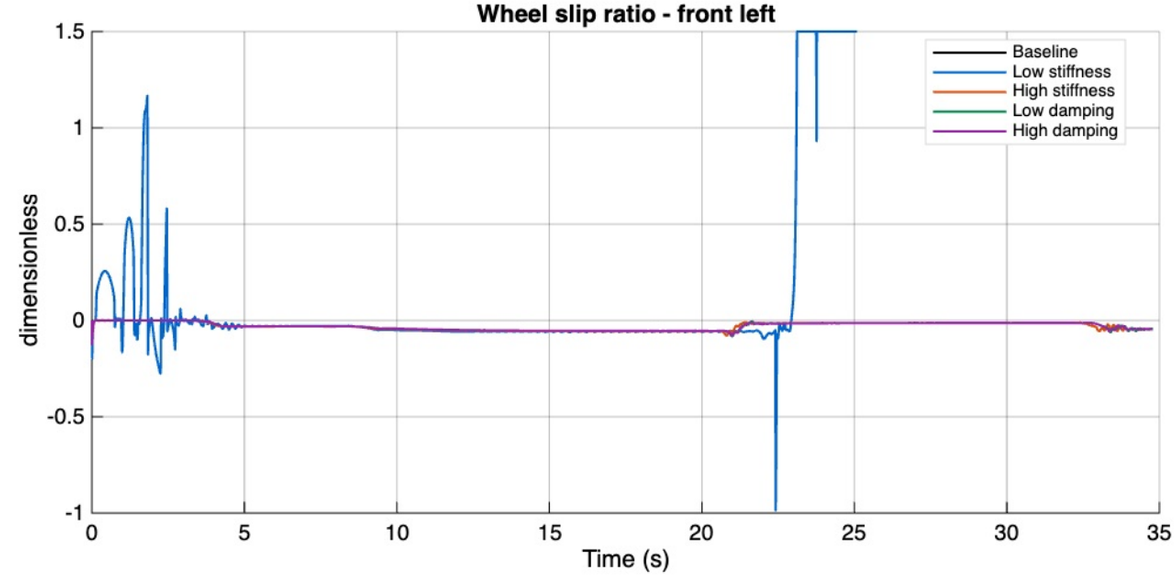


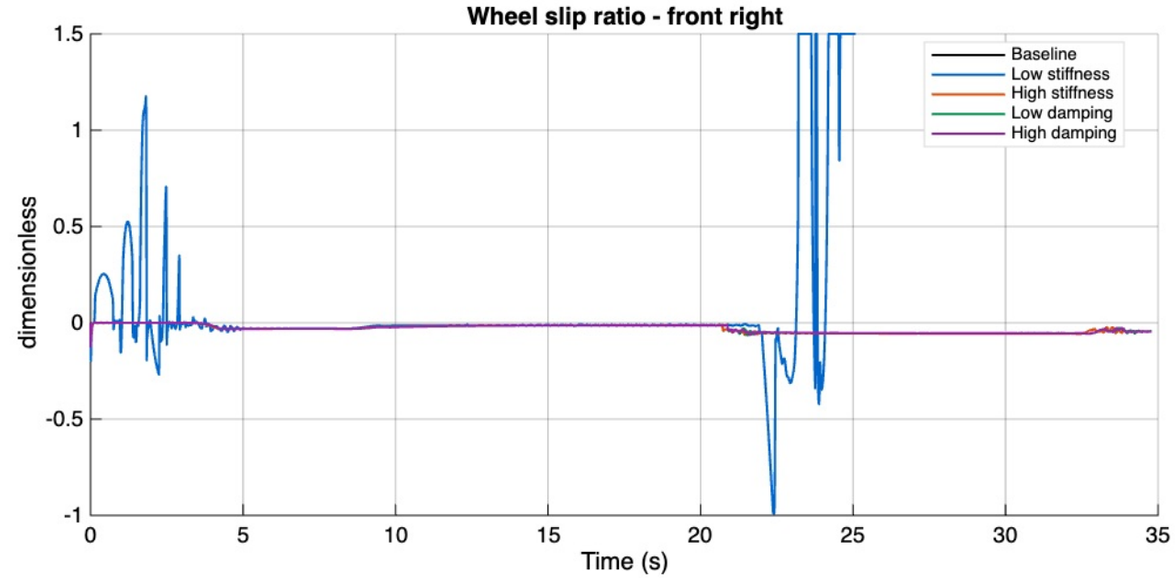
Suspension comparison - page 4

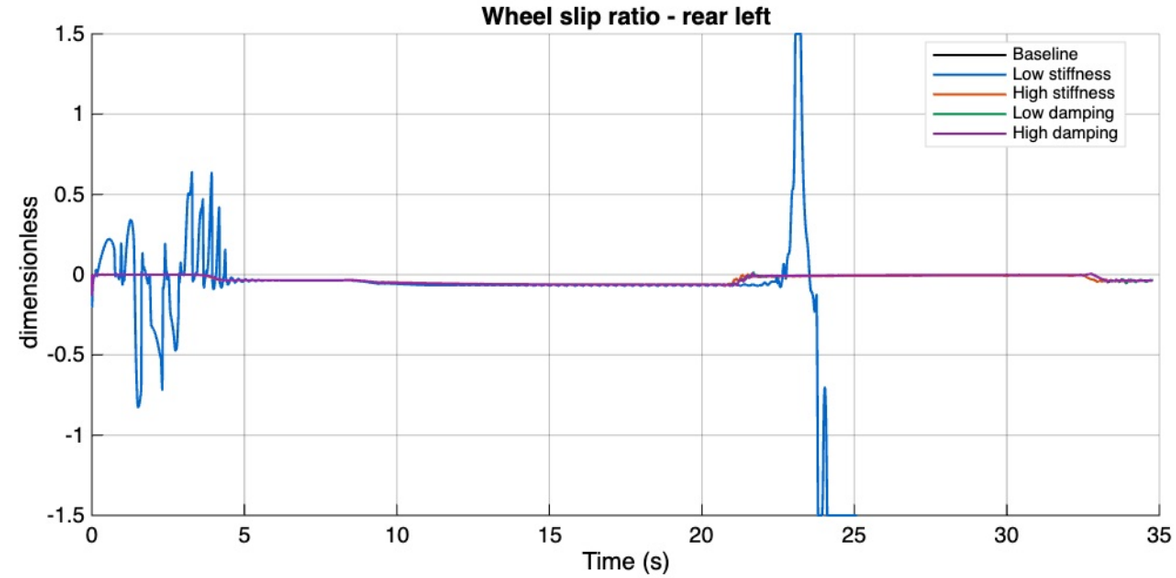


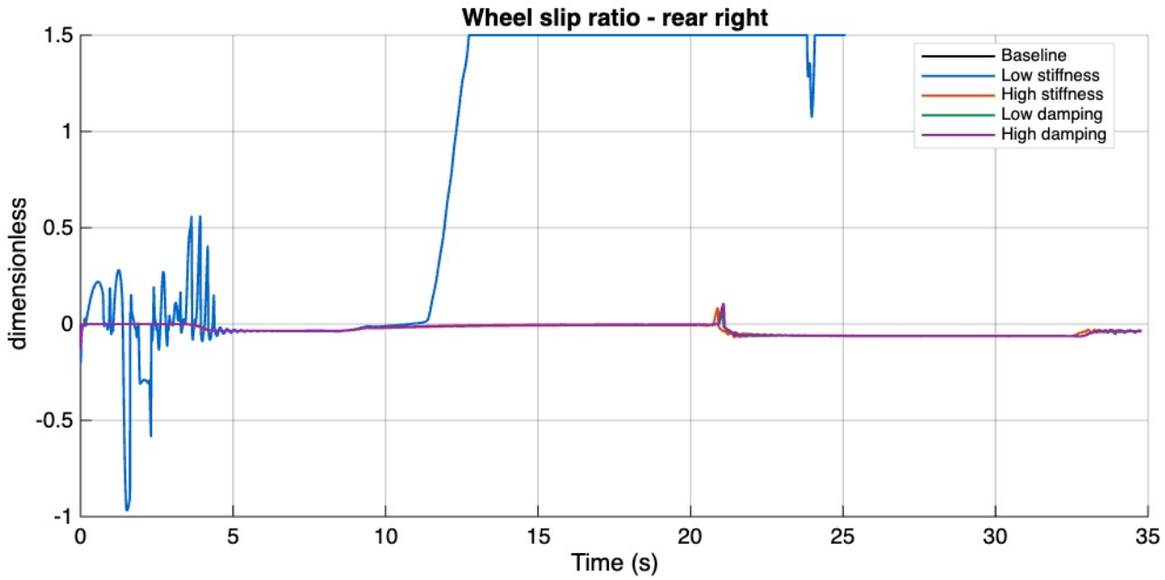


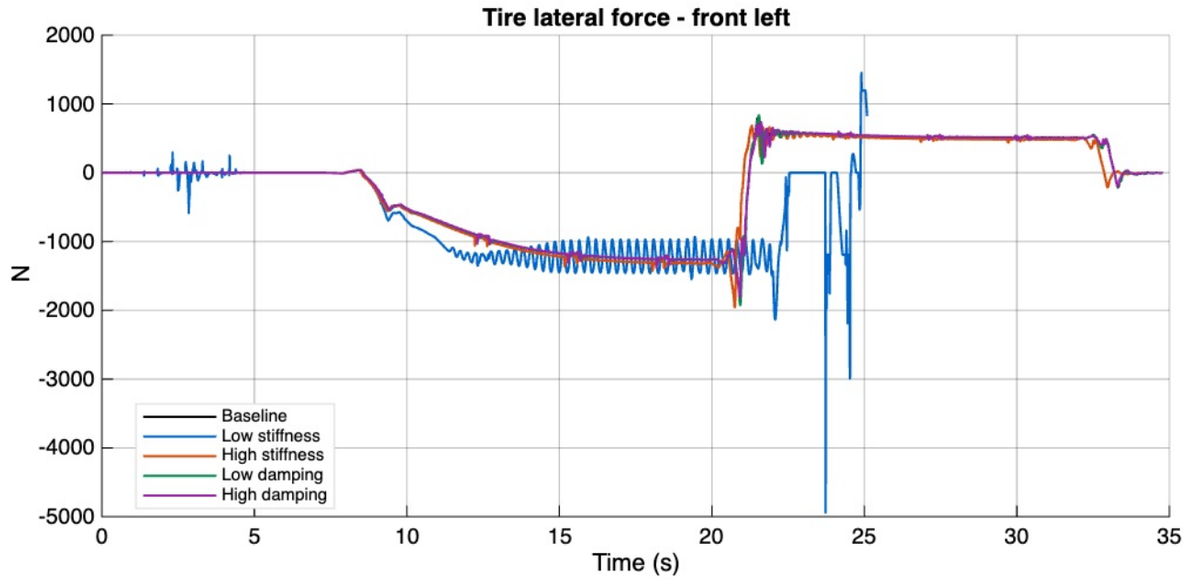


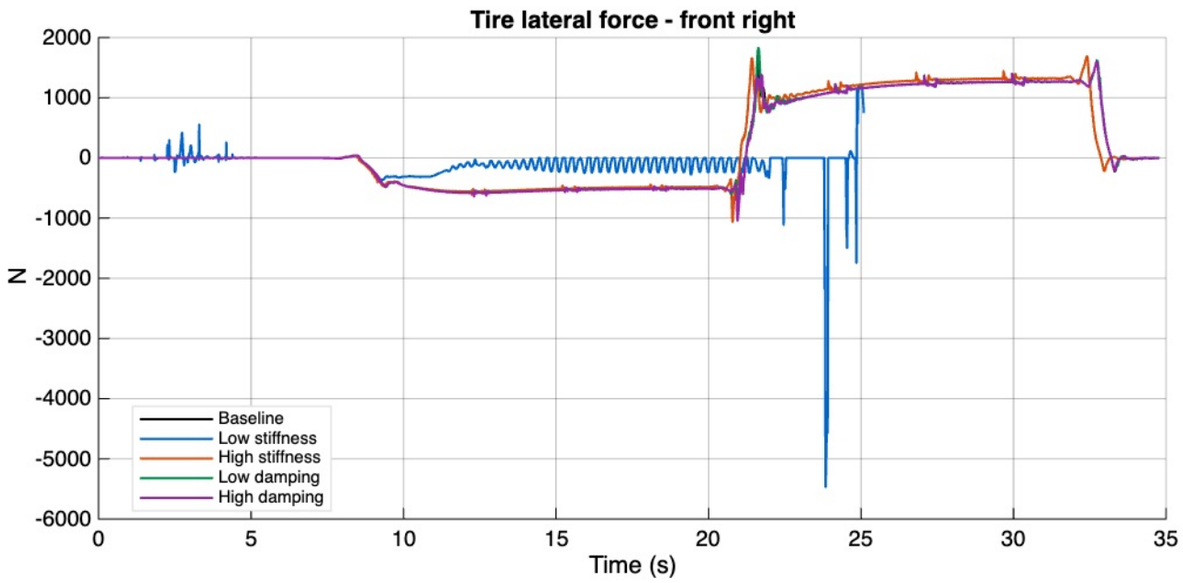


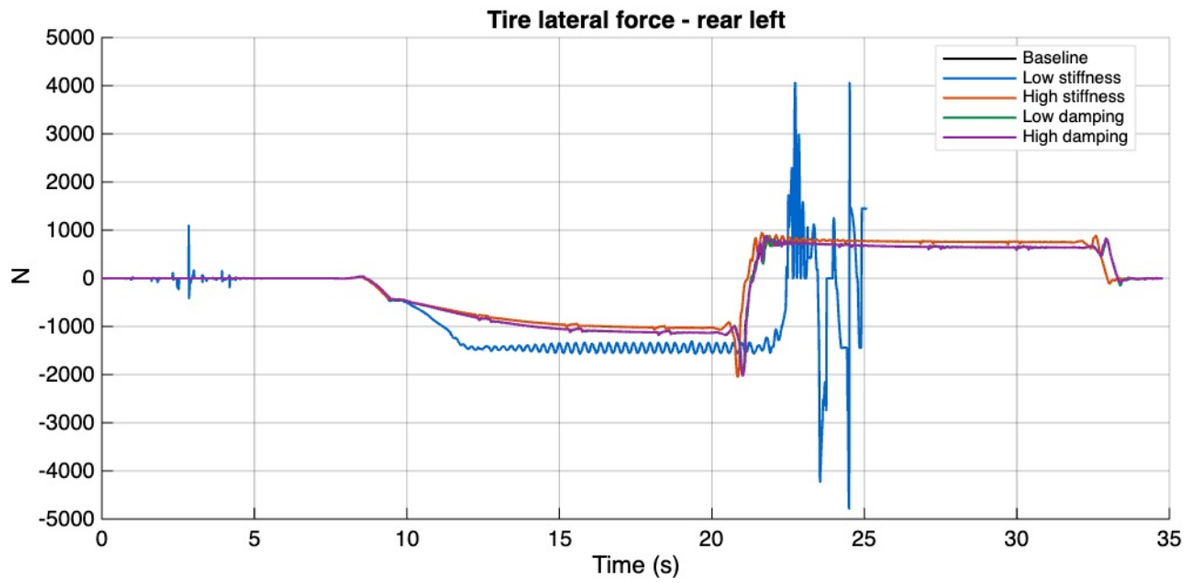


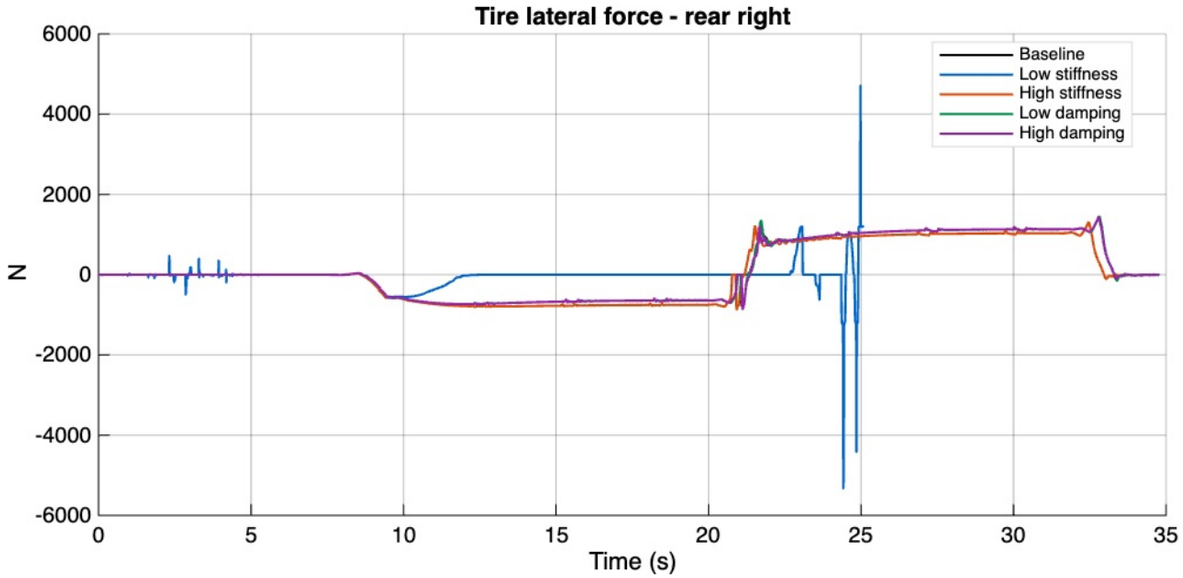


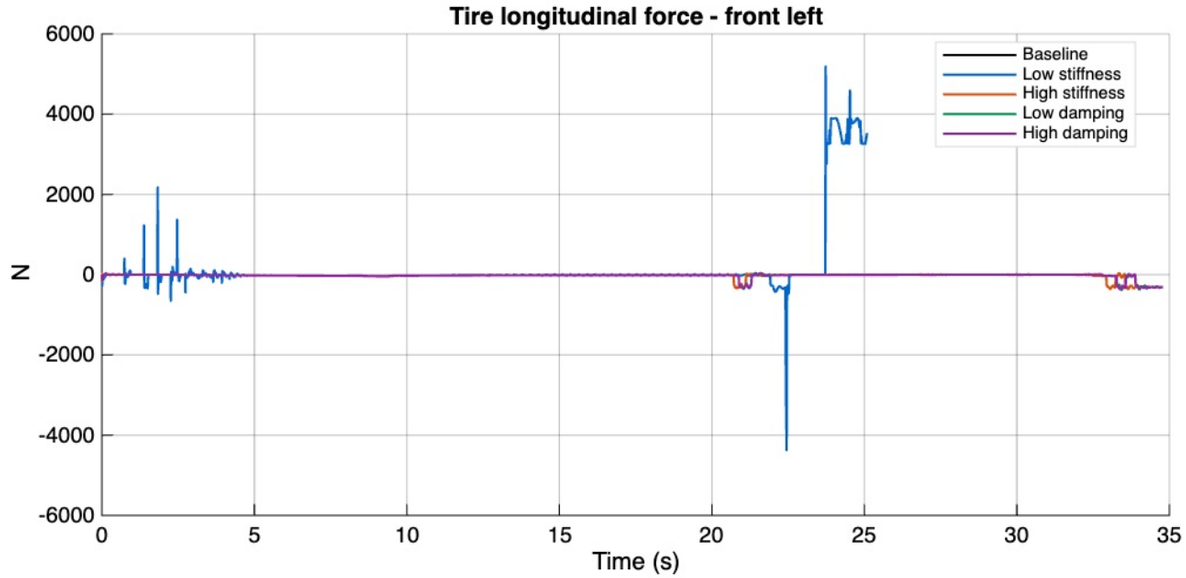

Wheel slip ratio - rear right
dimensionless
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

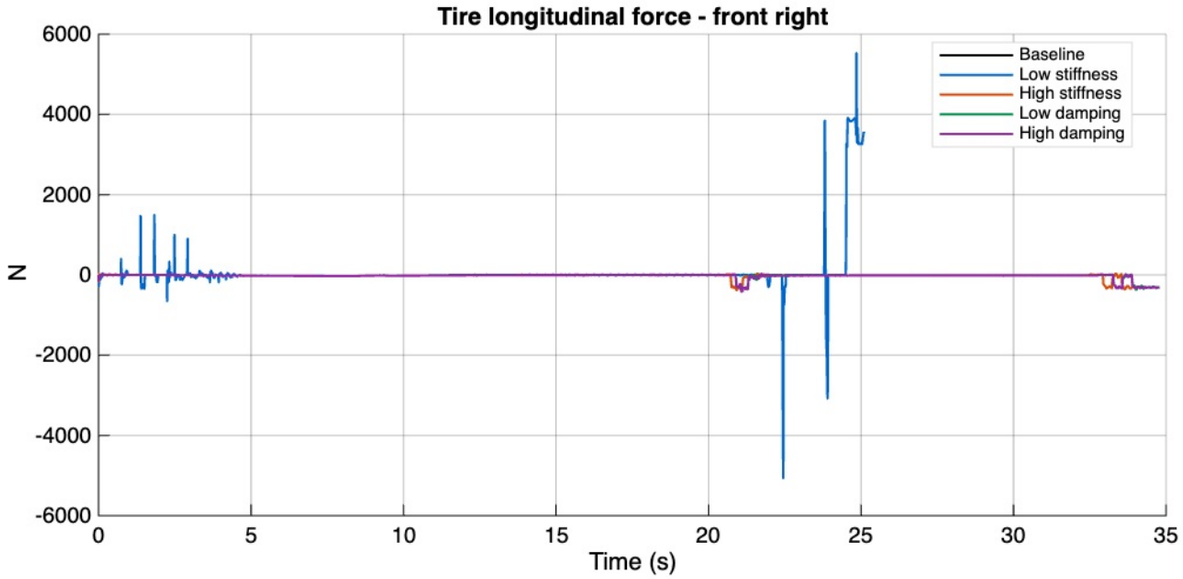

Tire lateral force - front left
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

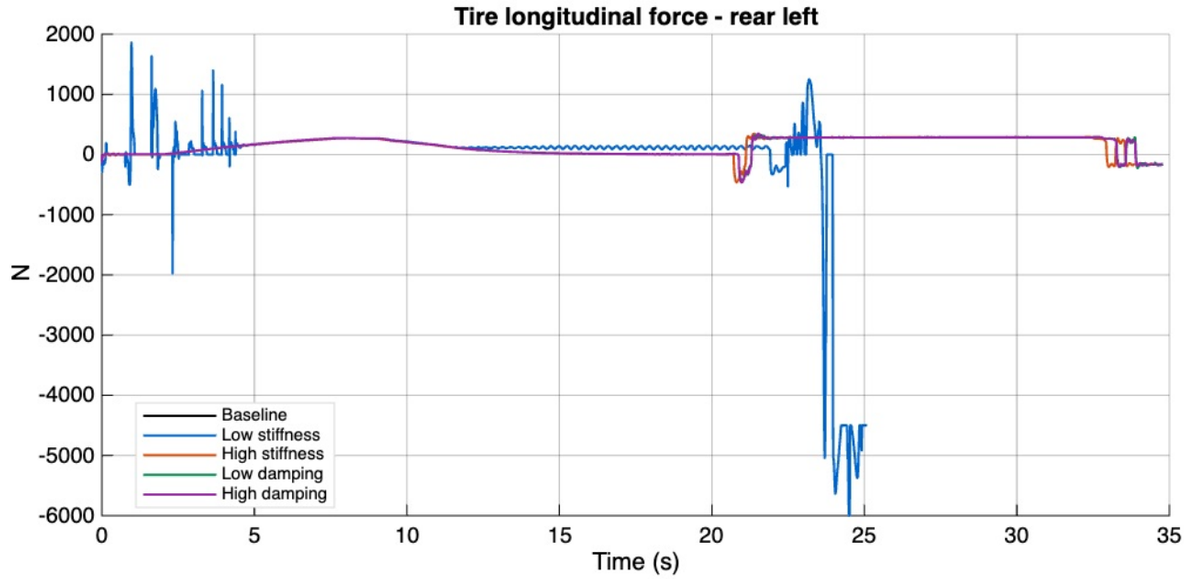

Tire lateral force - front right
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

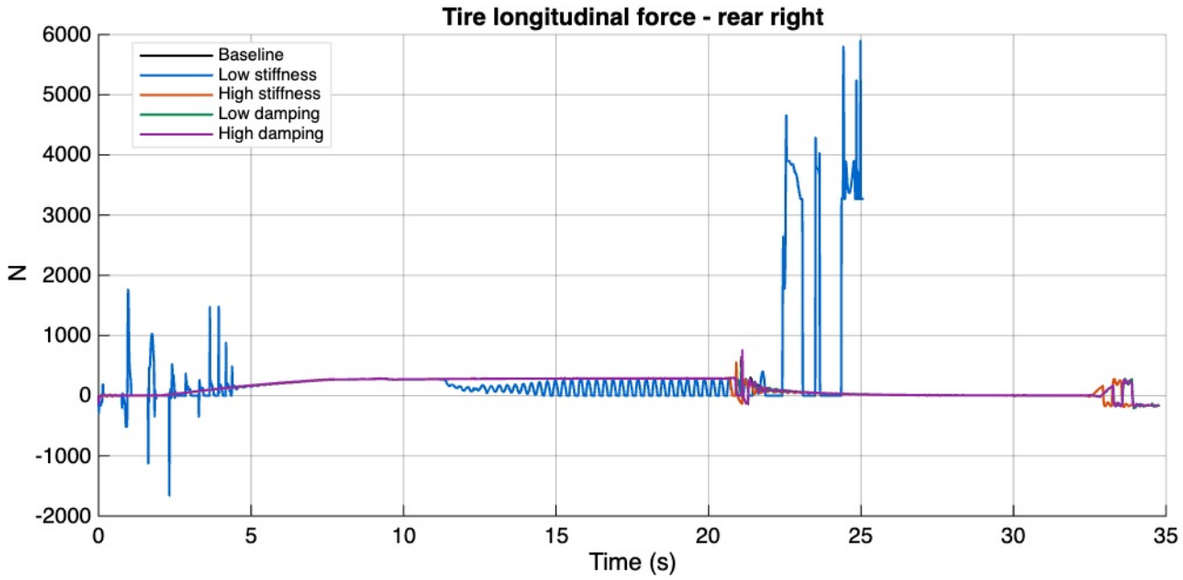

Tire lateral force - rear left
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

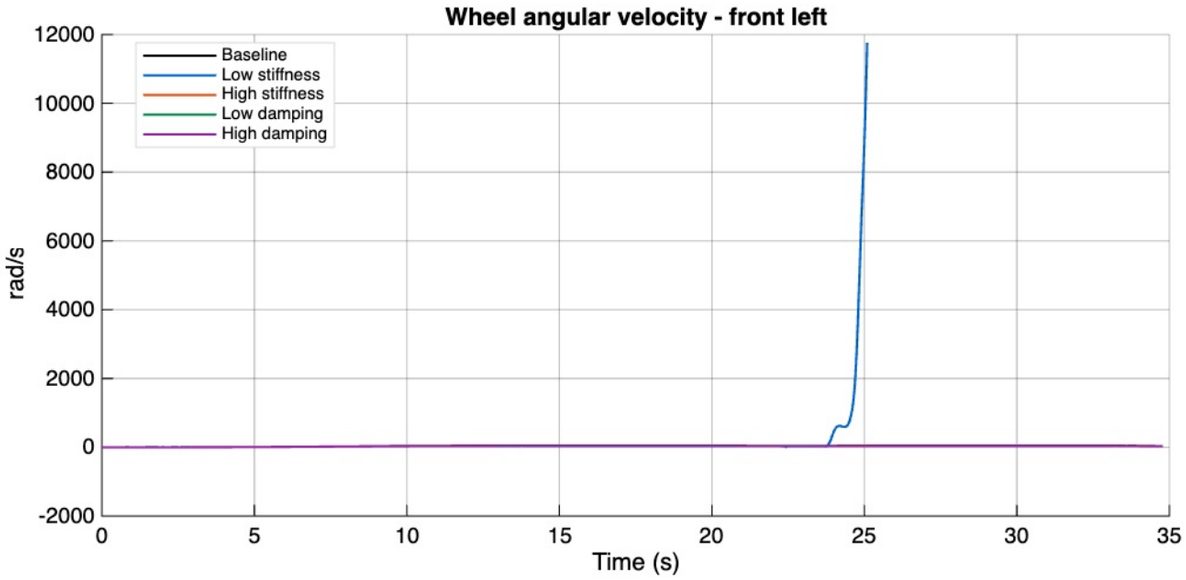

Tire lateral force - rear right
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

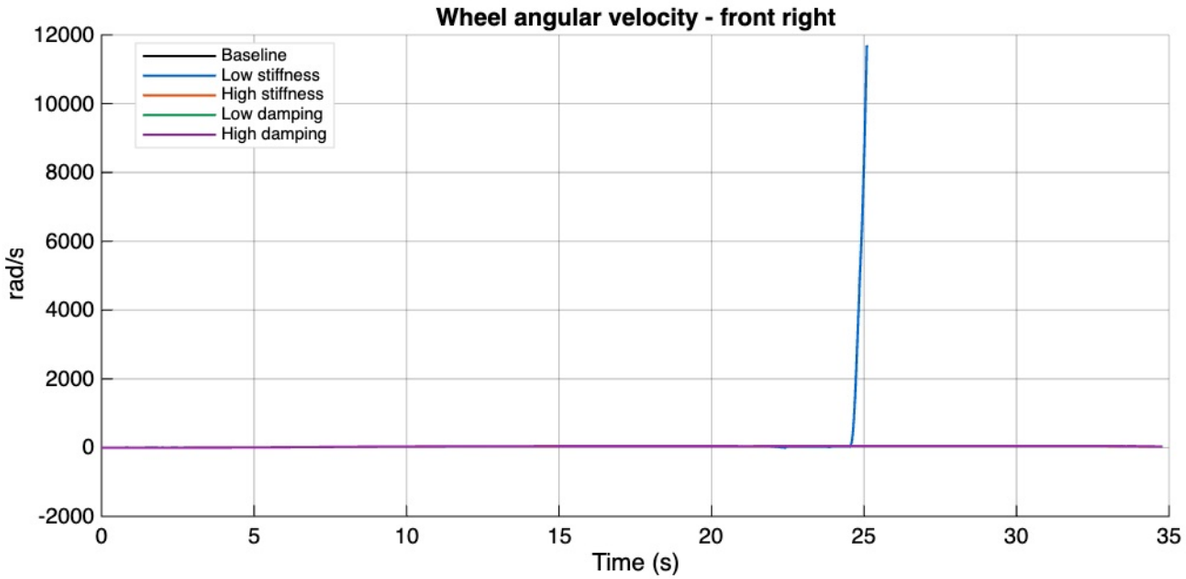

Tire longitudinal force - front left
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

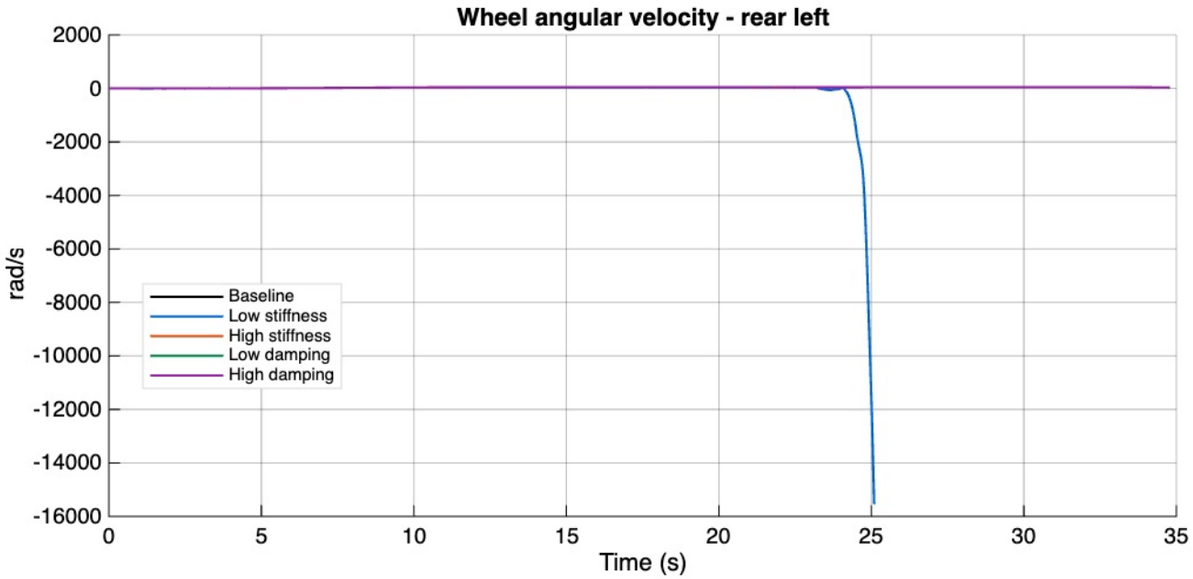

Tire longitudinal force - front right
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

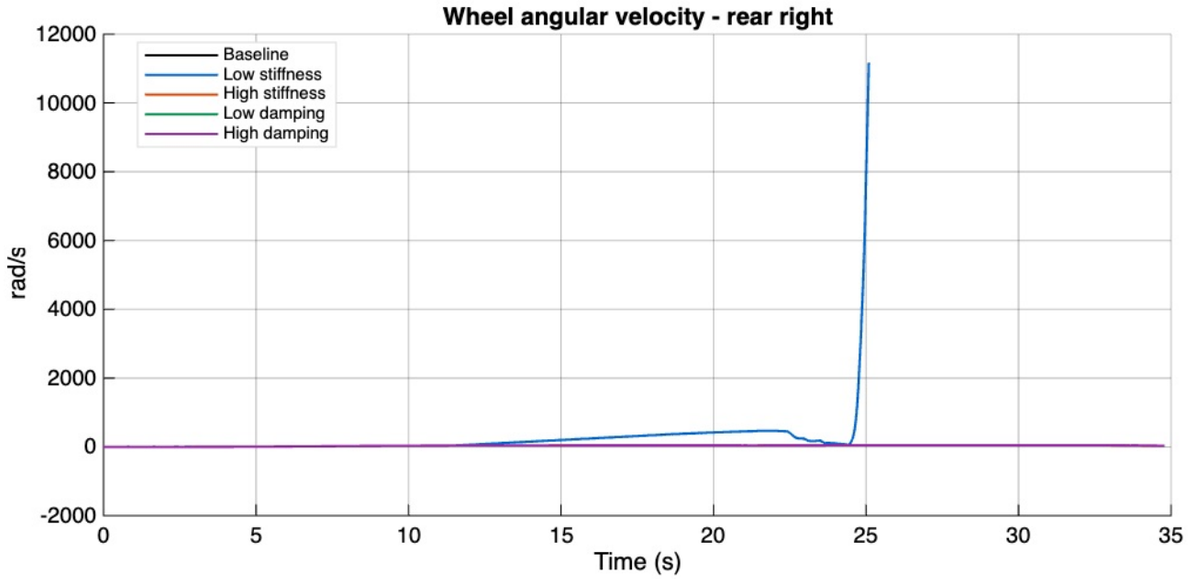

Tire longitudinal force - rear left
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

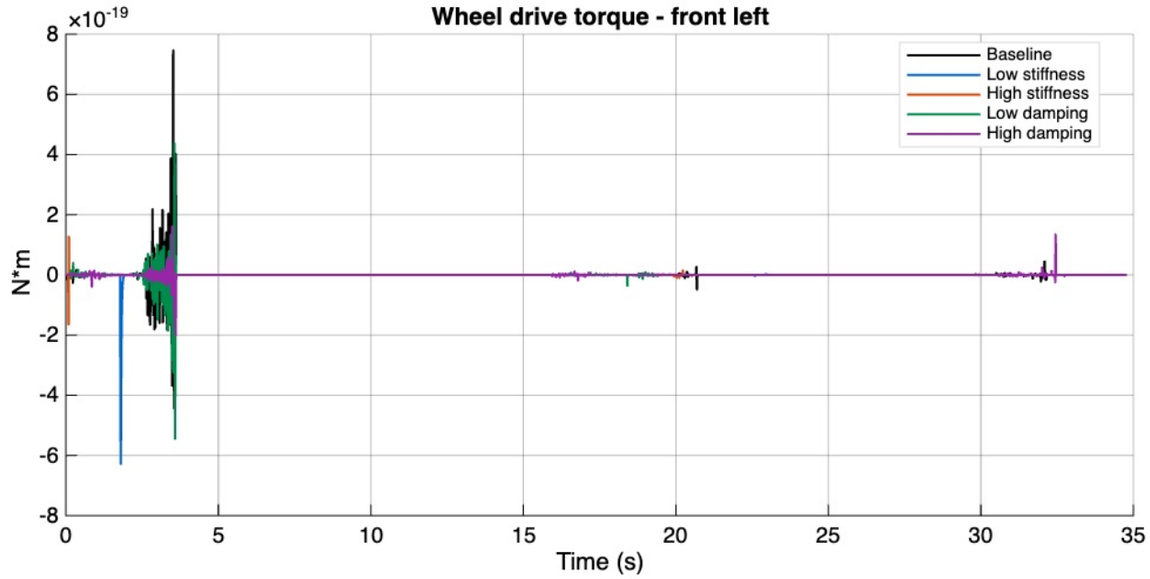

Tire longitudinal force - rear right
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

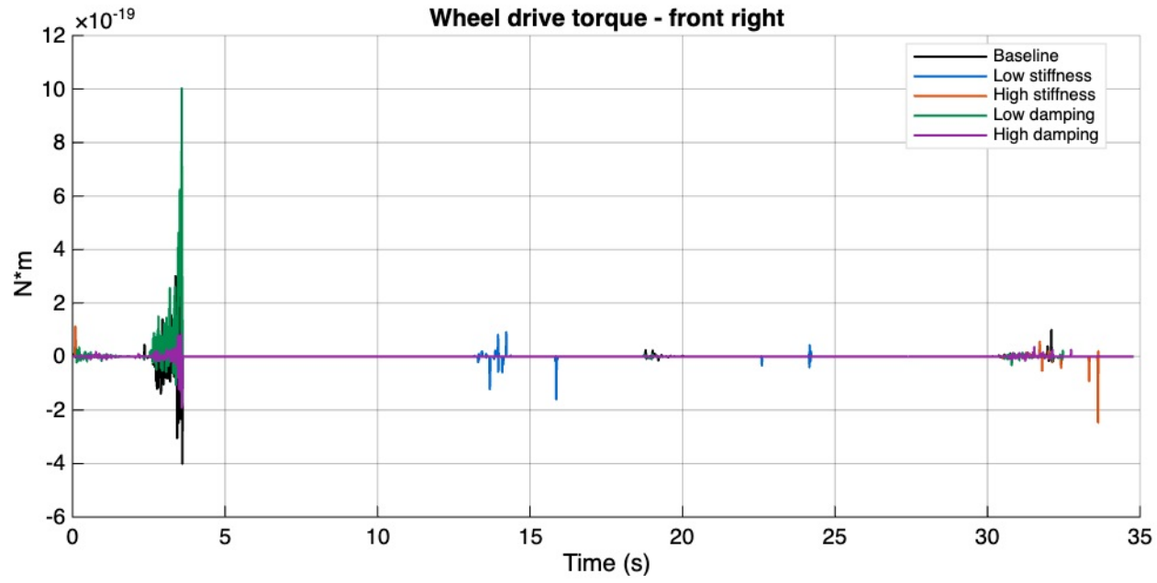

Wheel angular velocity - front left
rad/s
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

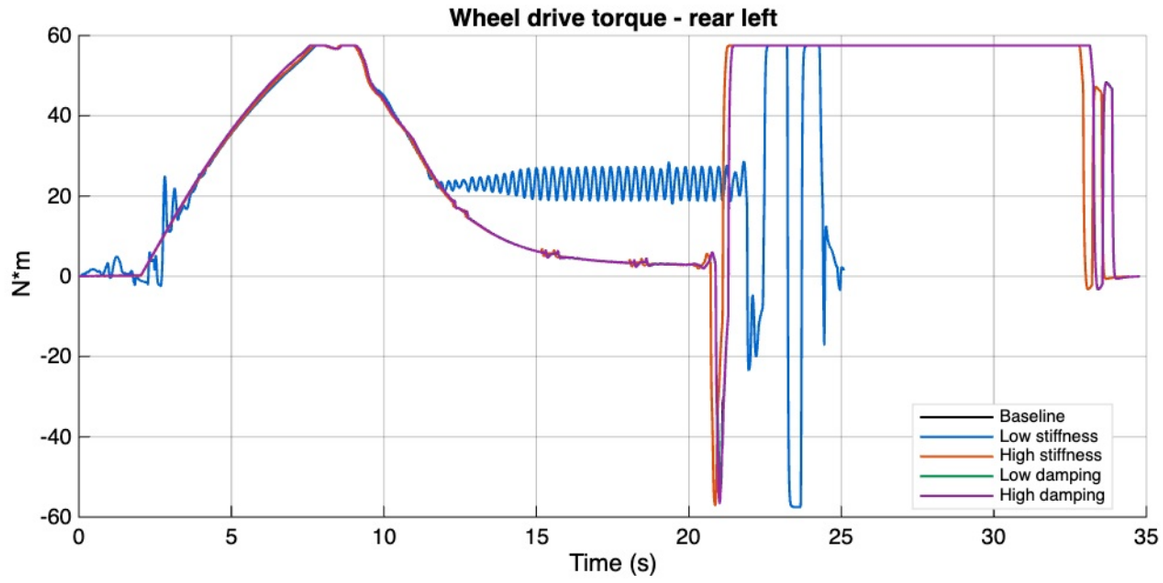

Wheel angular velocity - front right
rad/s
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

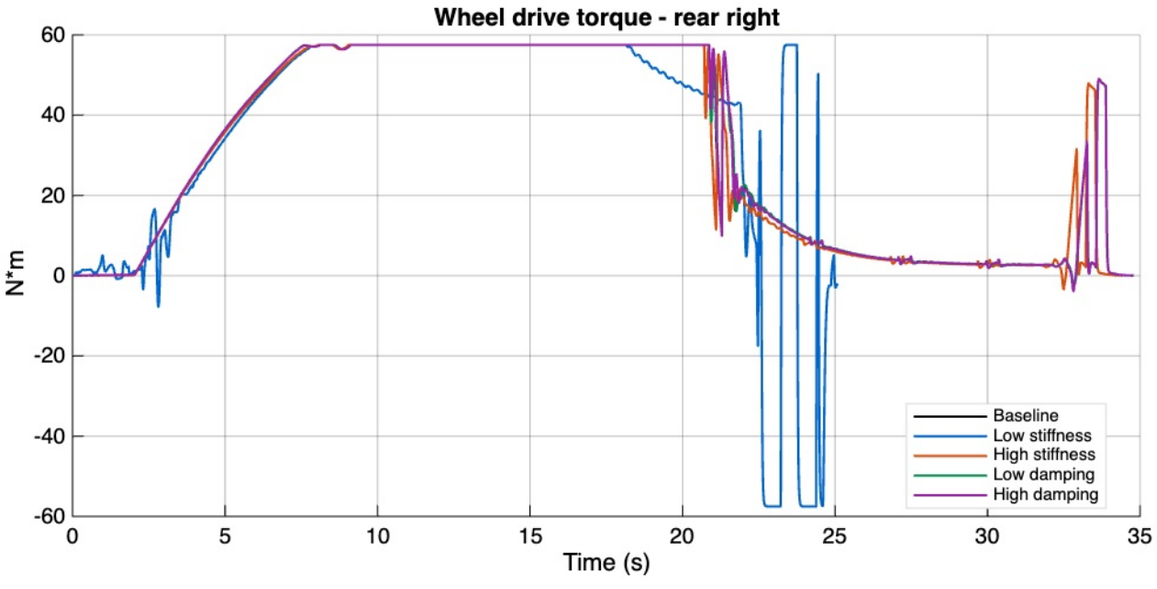

Wheel angular velocity - rear left
rad/s
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

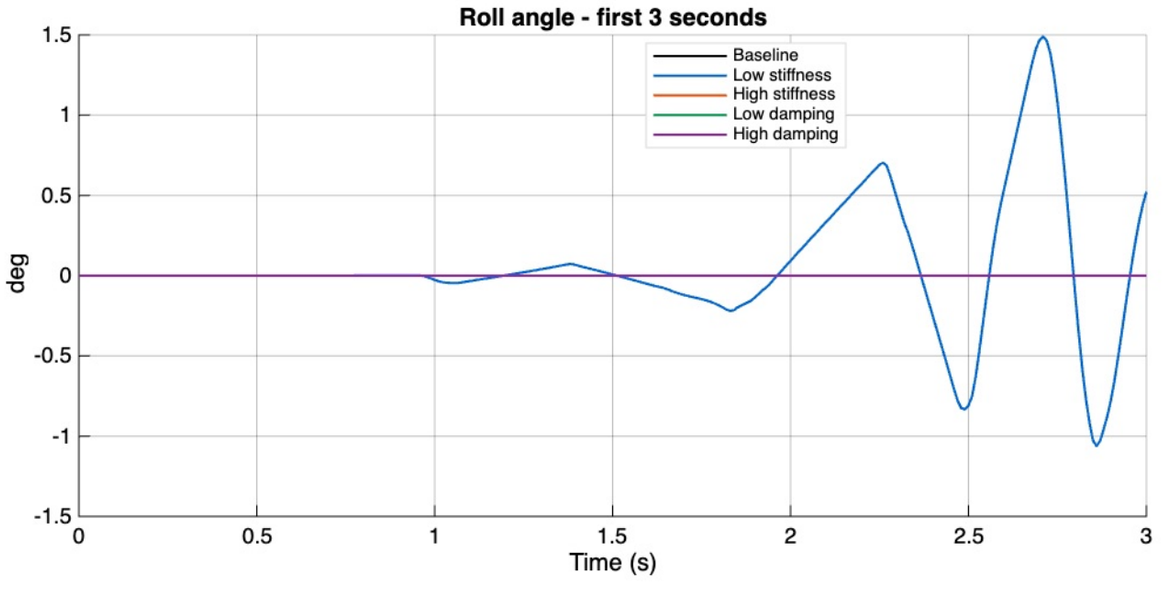

Wheel angular velocity - rear right
rad/s
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

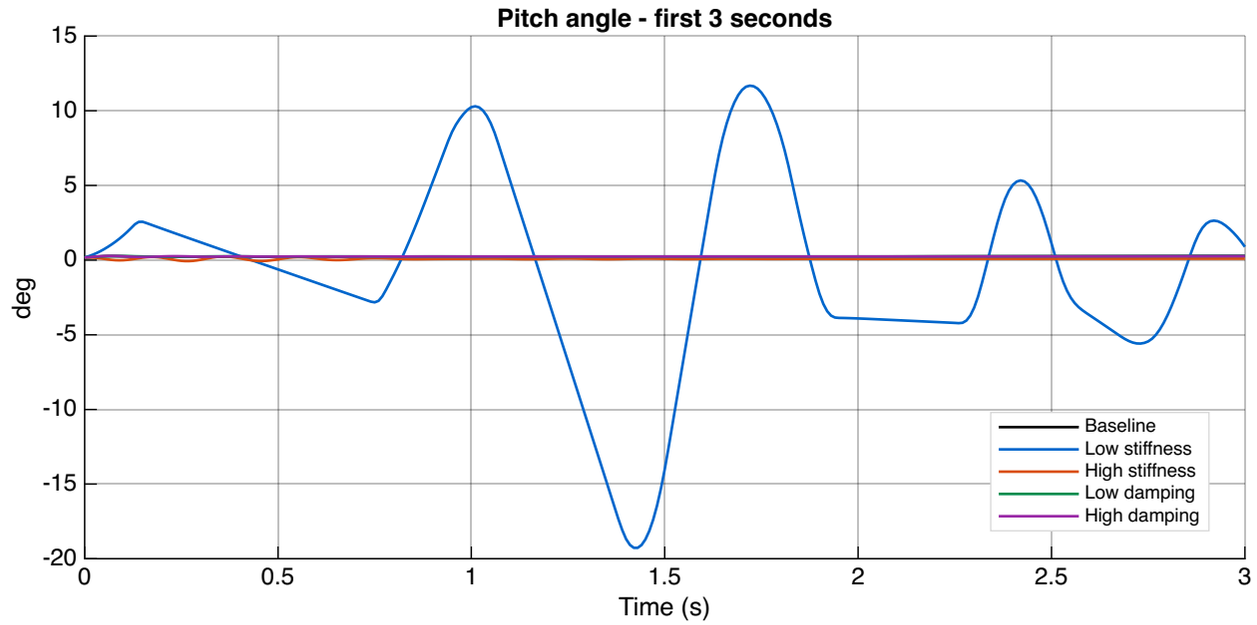

Wheel drive torque - front left
×10^-19
N*m
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

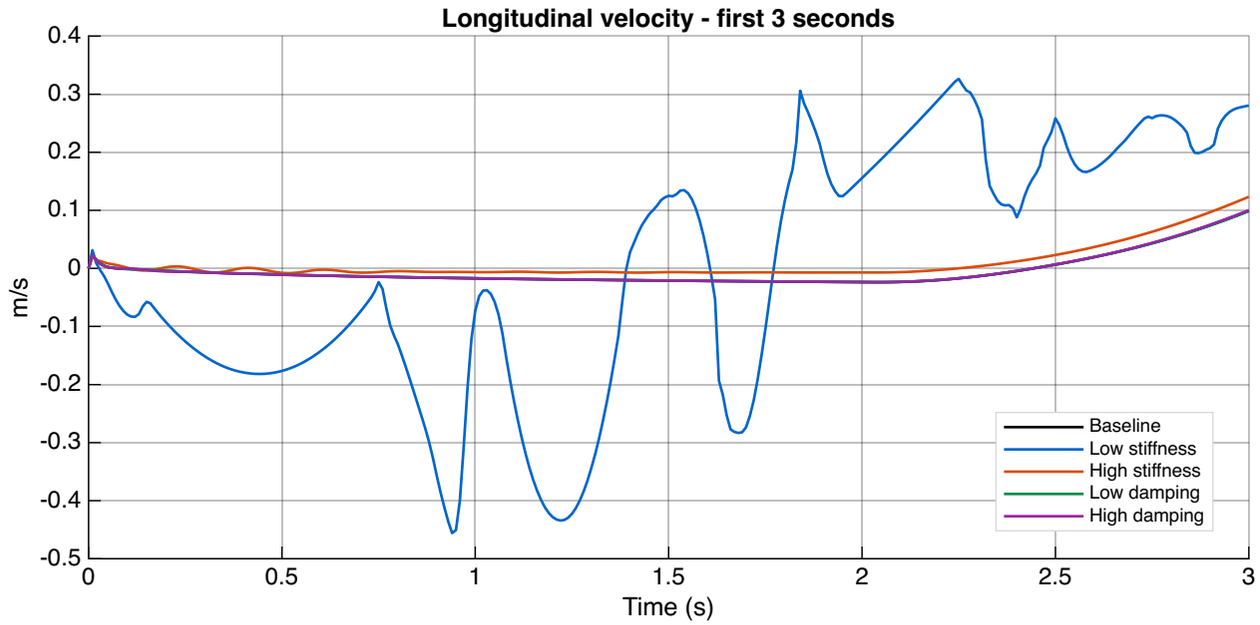

Wheel drive torque - front right
×10^-19
N*m
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

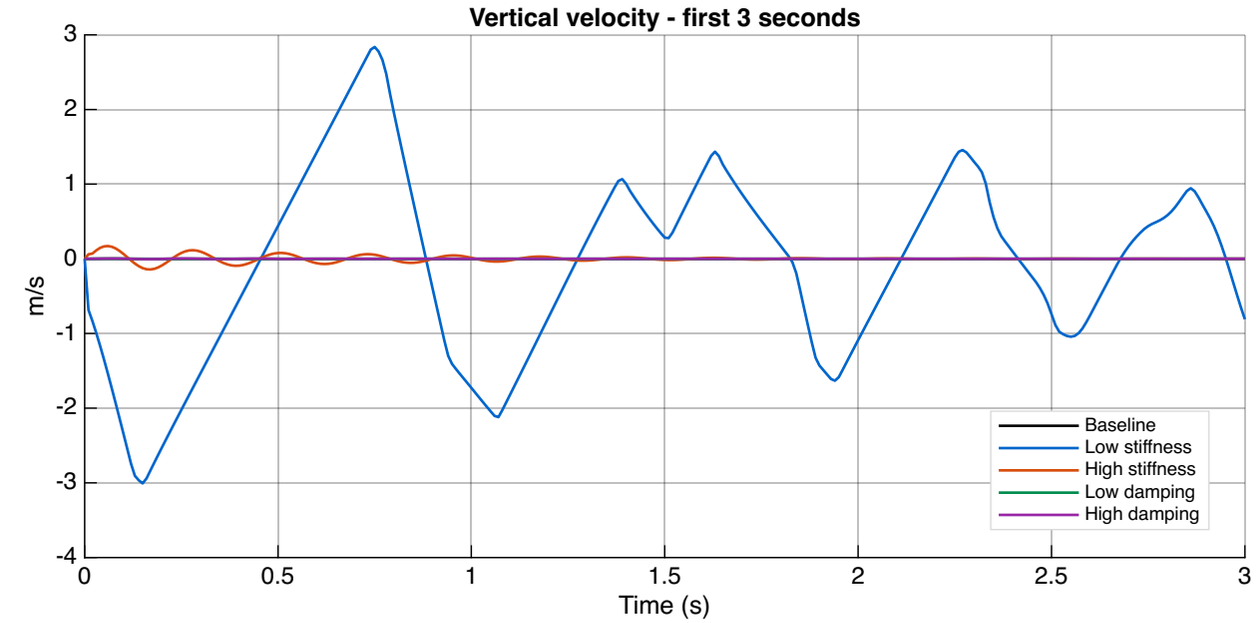

Wheel drive torque - rear left
N*m
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

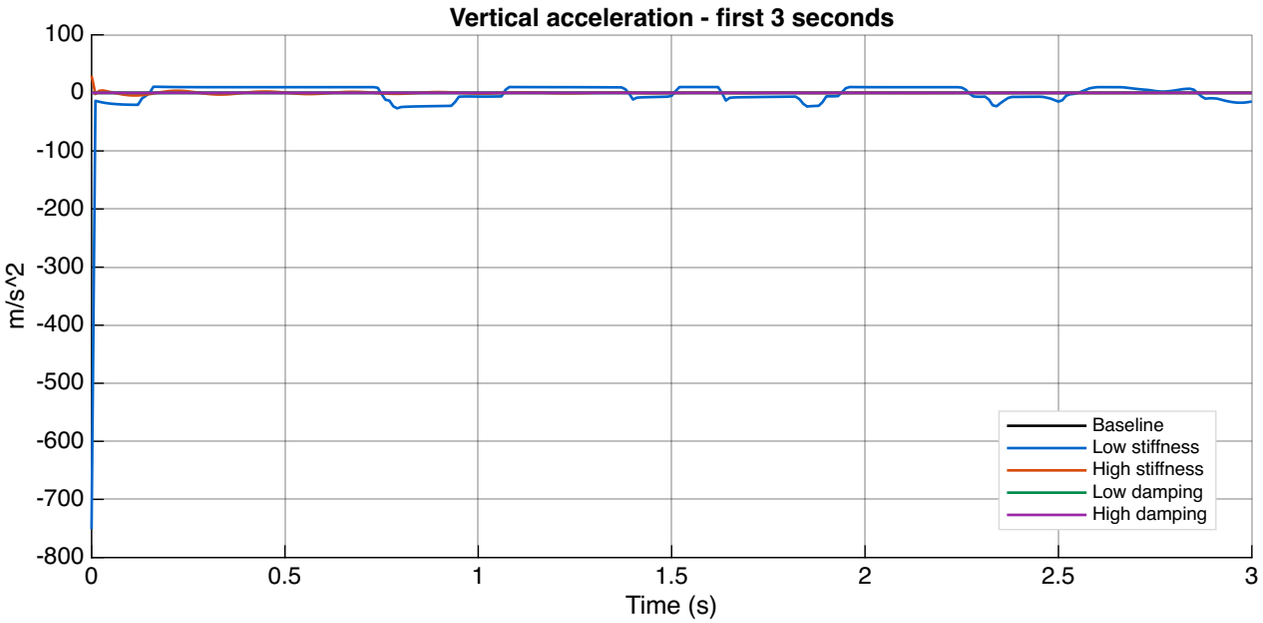

Wheel drive torque - rear right
N*m
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

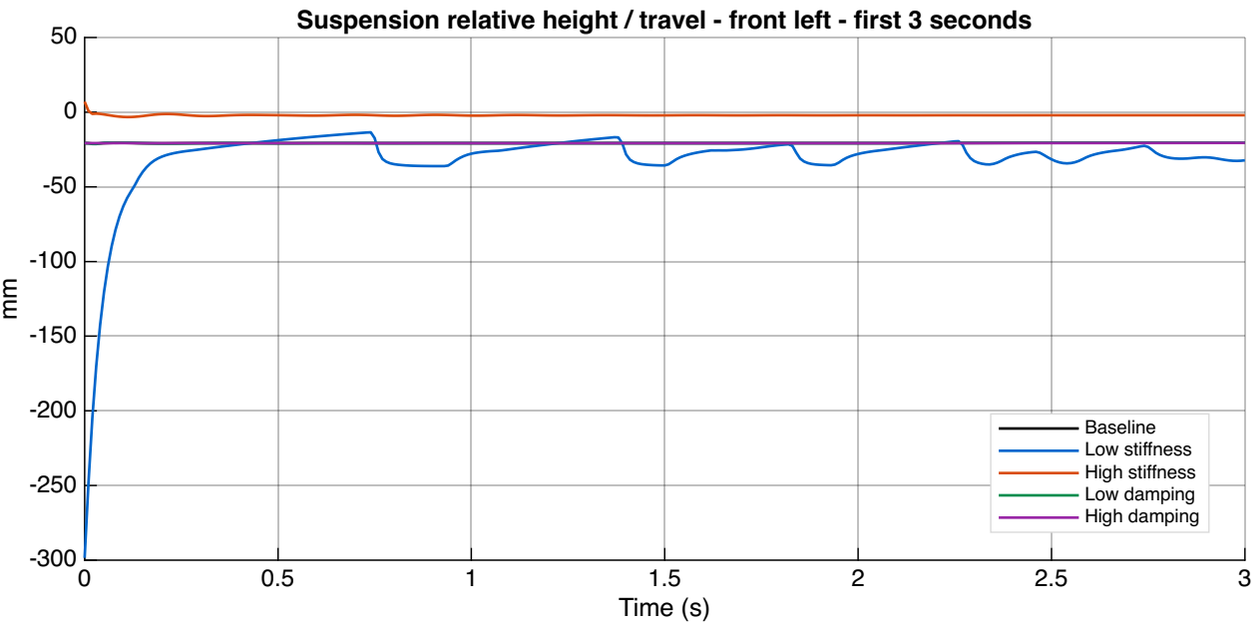

Roll angle - first 3 seconds
deg
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

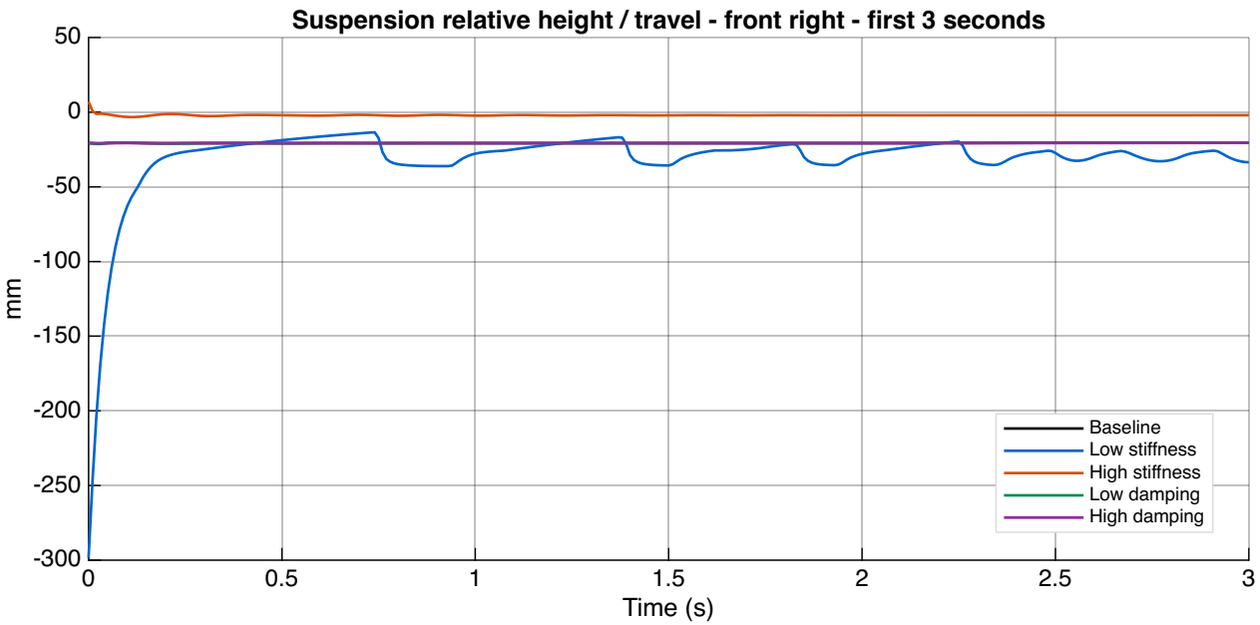

Pitch angle - first 3 seconds
deg
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

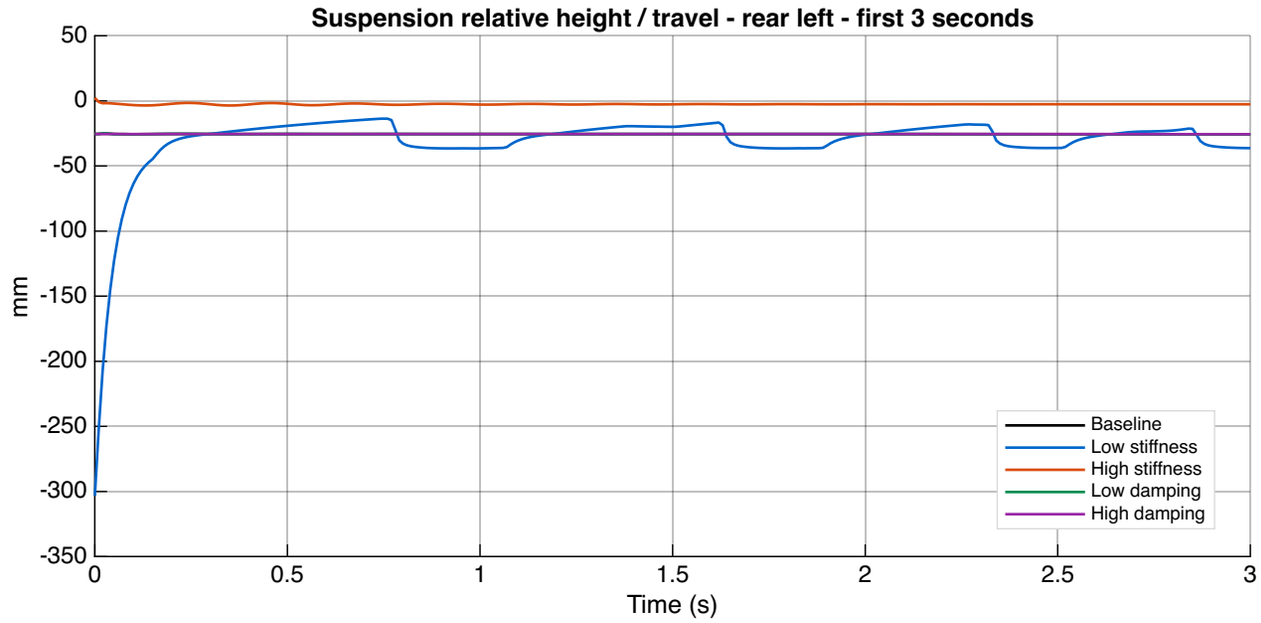

Longitudinal velocity - first 3 seconds
m/s
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

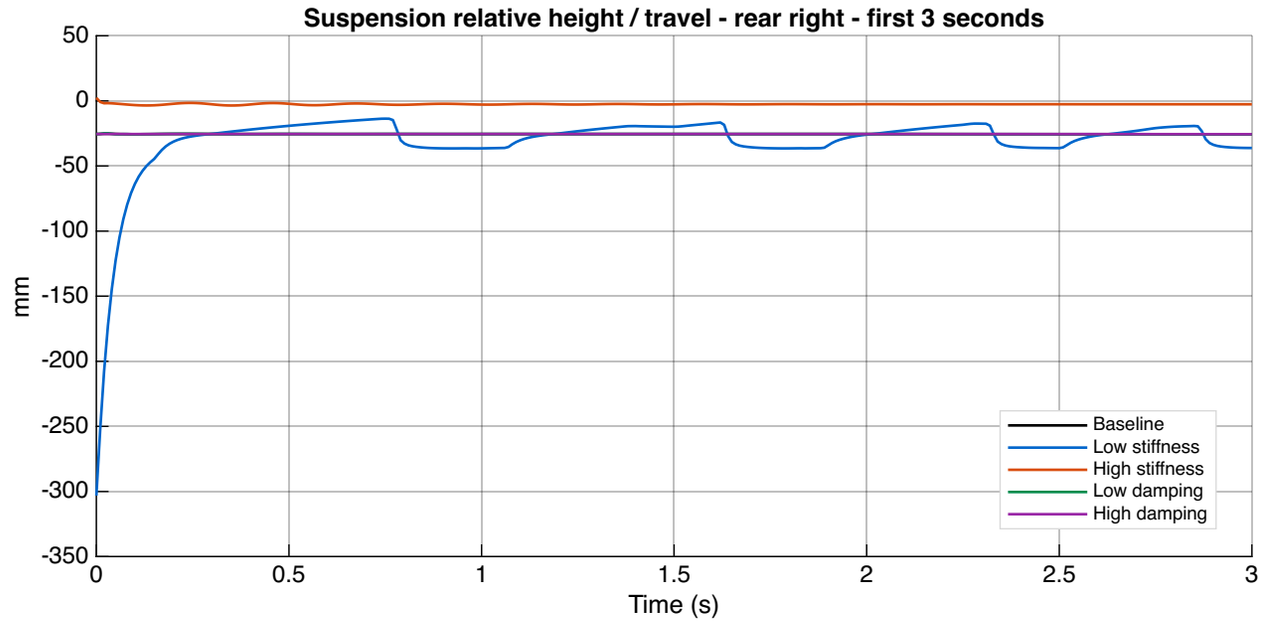

Vertical velocity - first 3 seconds
m/s
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

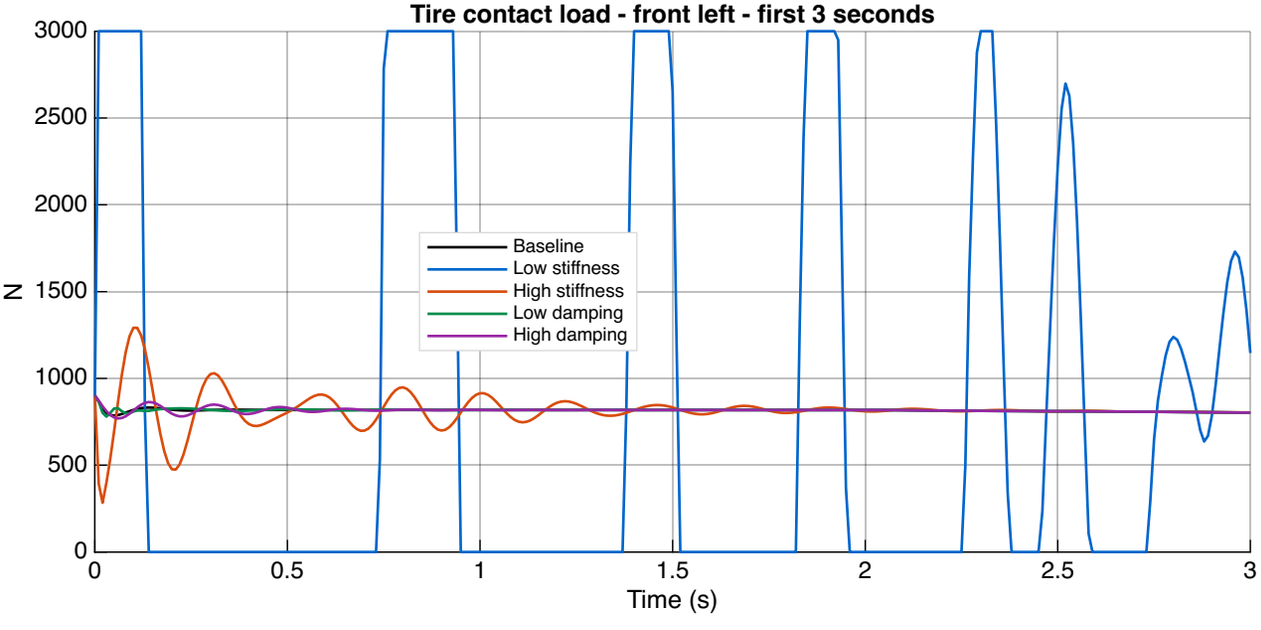

Vertical acceleration - first 3 seconds
m/s^2
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

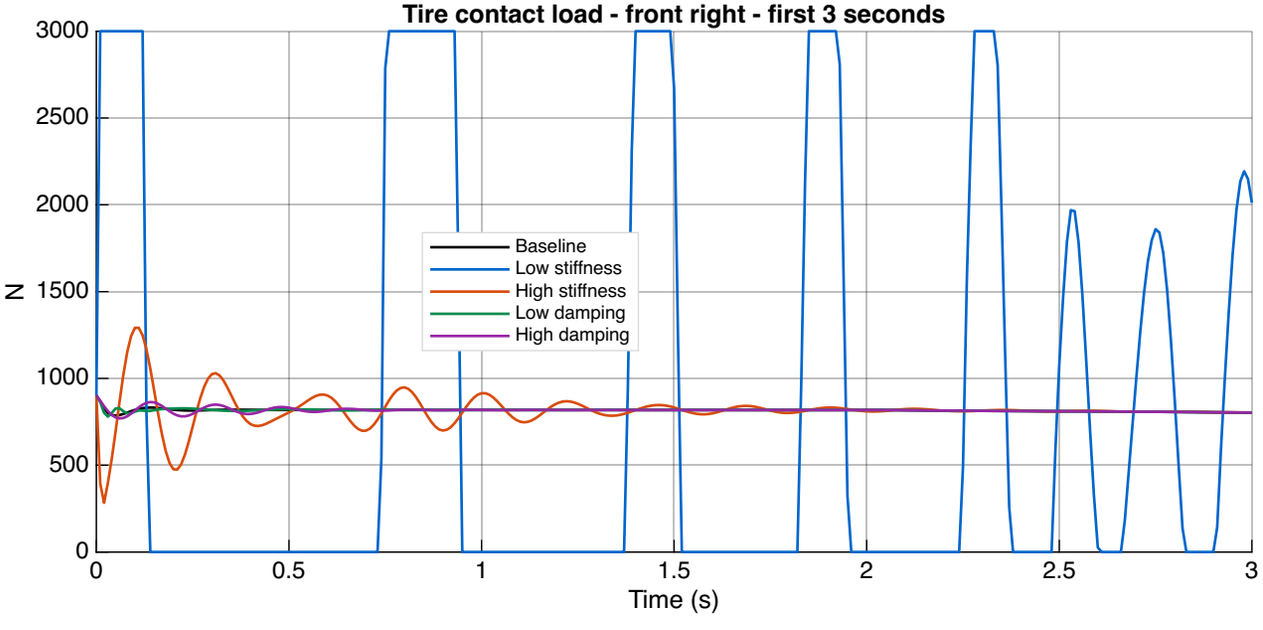

Suspension relative height / travel - front left - first 3 seconds
mm
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

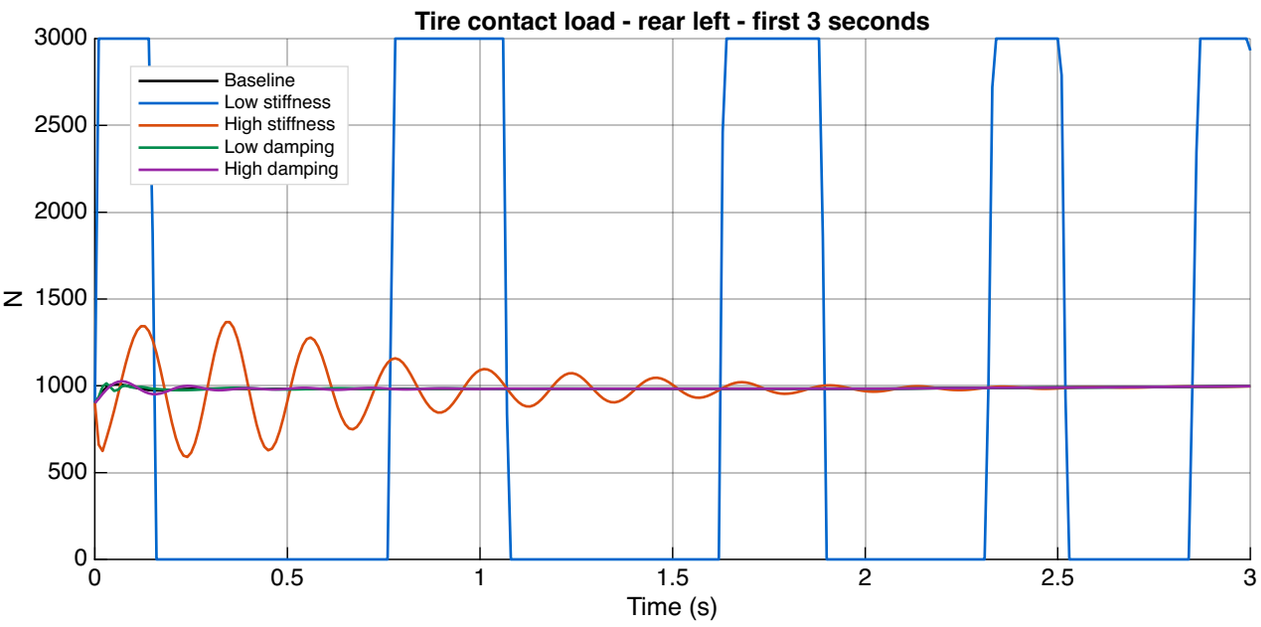

Suspension relative height / travel - front right - first 3 seconds
mm
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

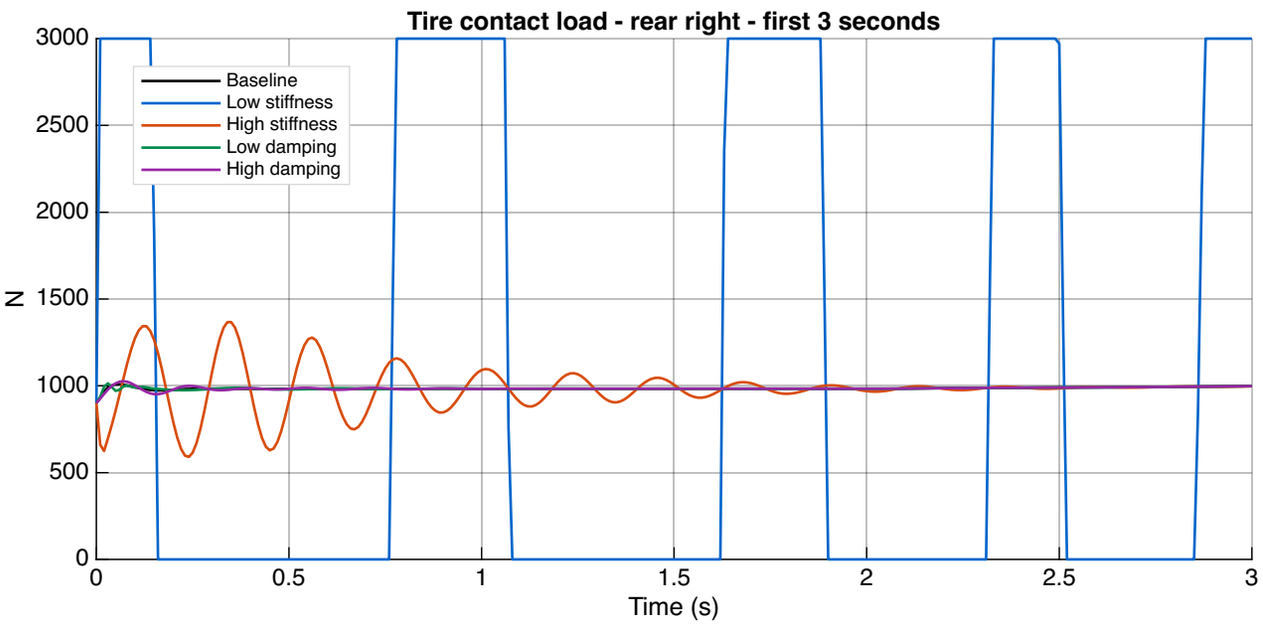

Suspension relative height / travel - rear left - first 3 seconds
mm
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

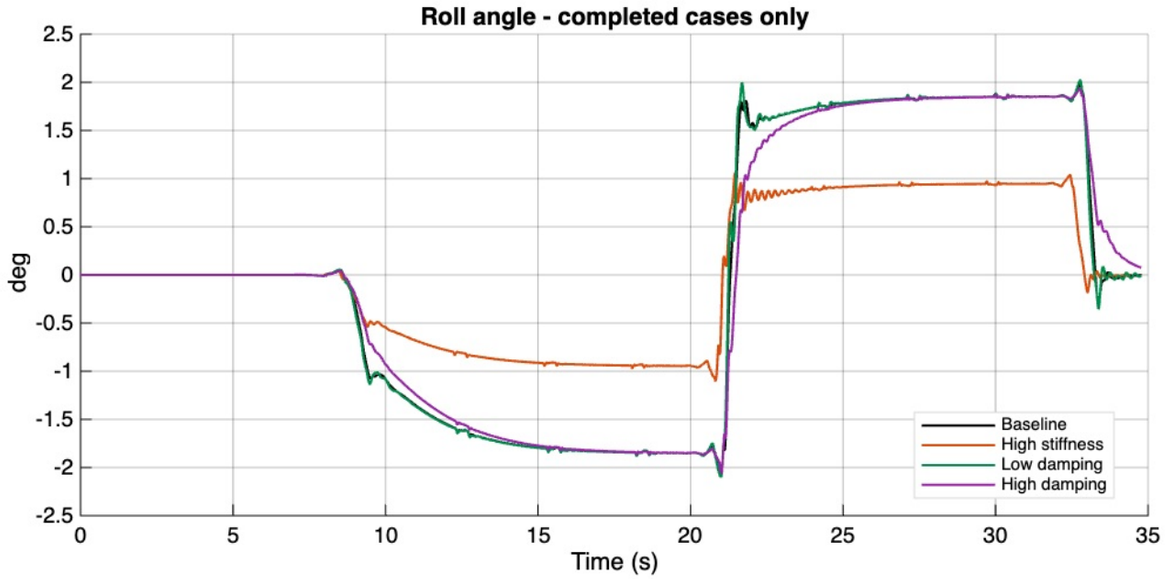

Suspension relative height / travel - rear right - first 3 seconds
mm
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

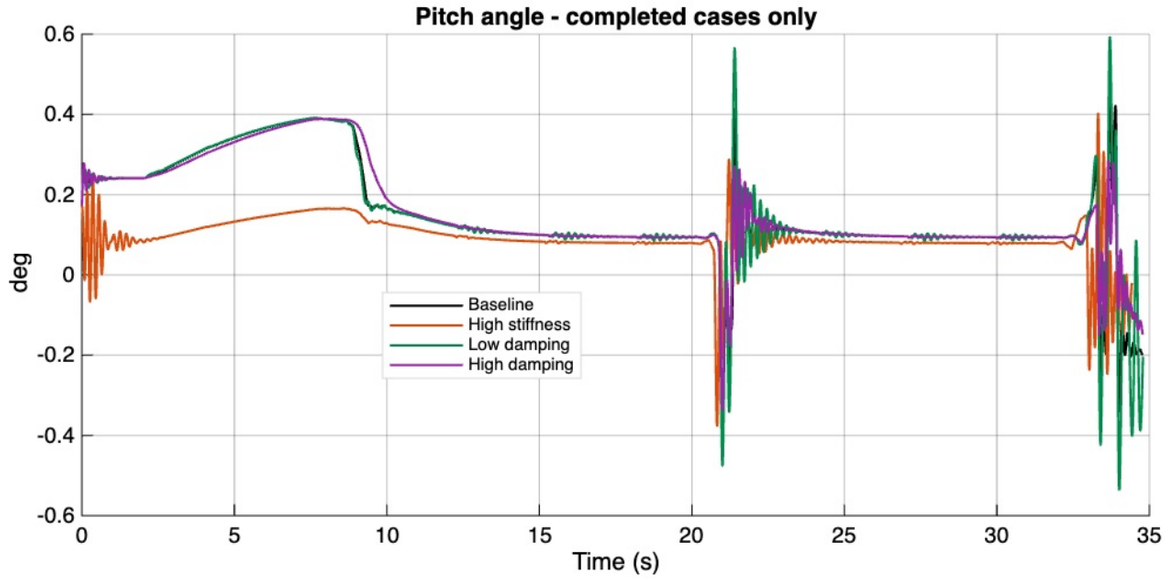

Tire contact load - front left - first 3 seconds
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

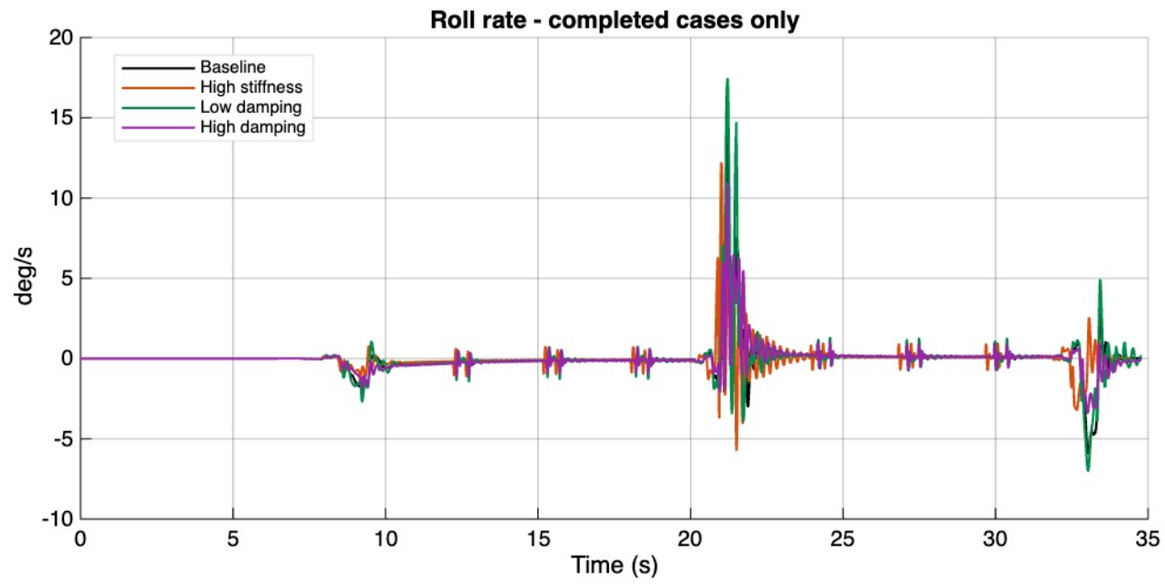

Tire contact load - front right - first 3 seconds
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

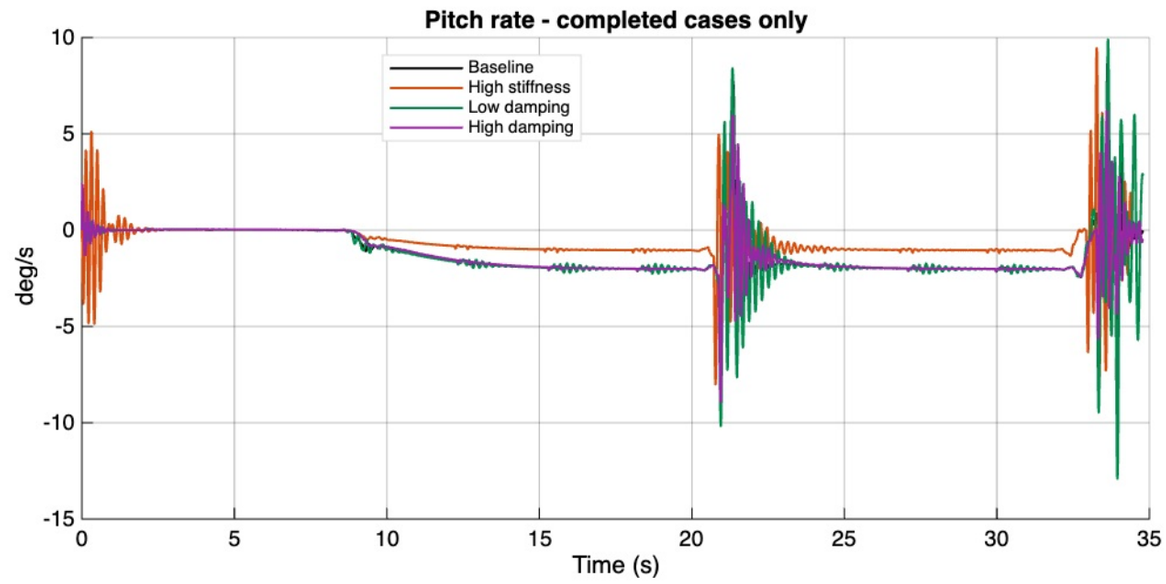

Tire contact load - rear left - first 3 seconds
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

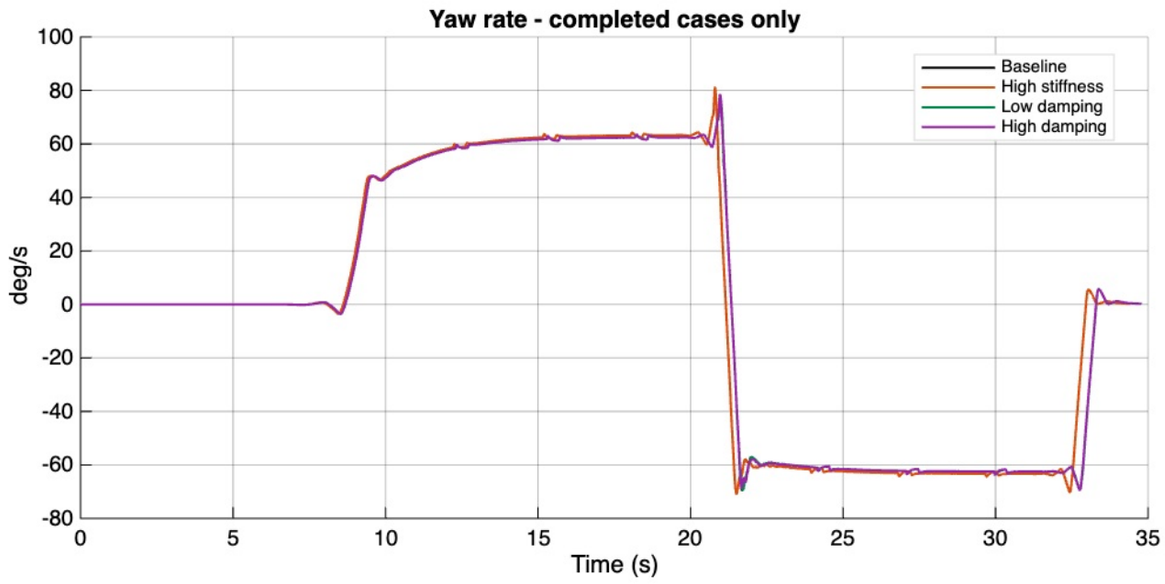

Tire contact load - rear right - first 3 seconds
N
Time (s)
Baseline
Low stiffness
High stiffness
Low damping
High damping

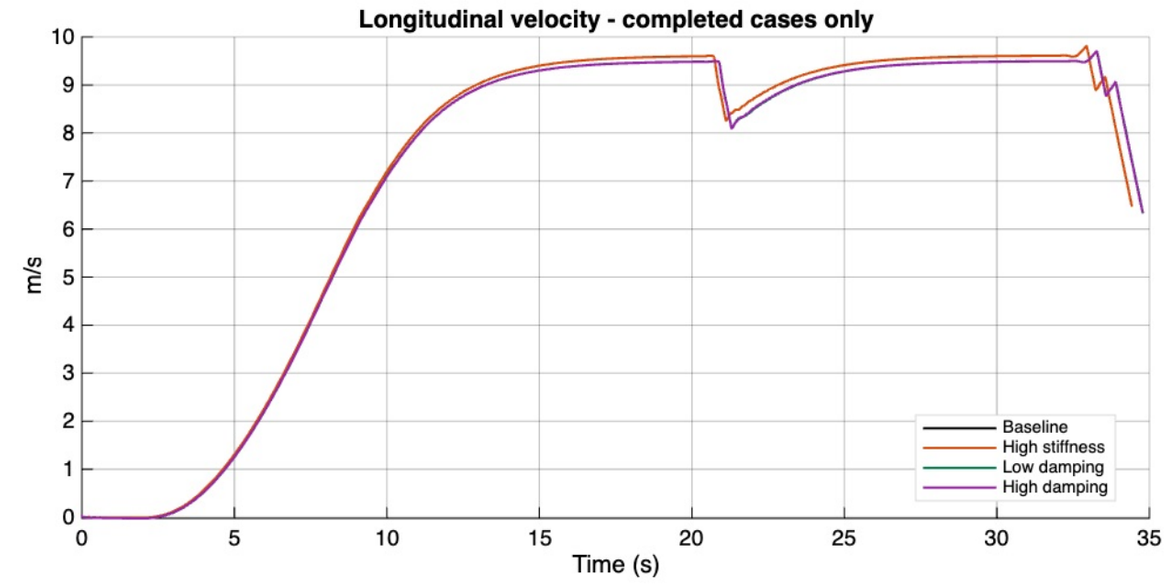

Roll angle - completed cases only
deg
Time (s)
Baseline
High stiffness
Low damping
High damping

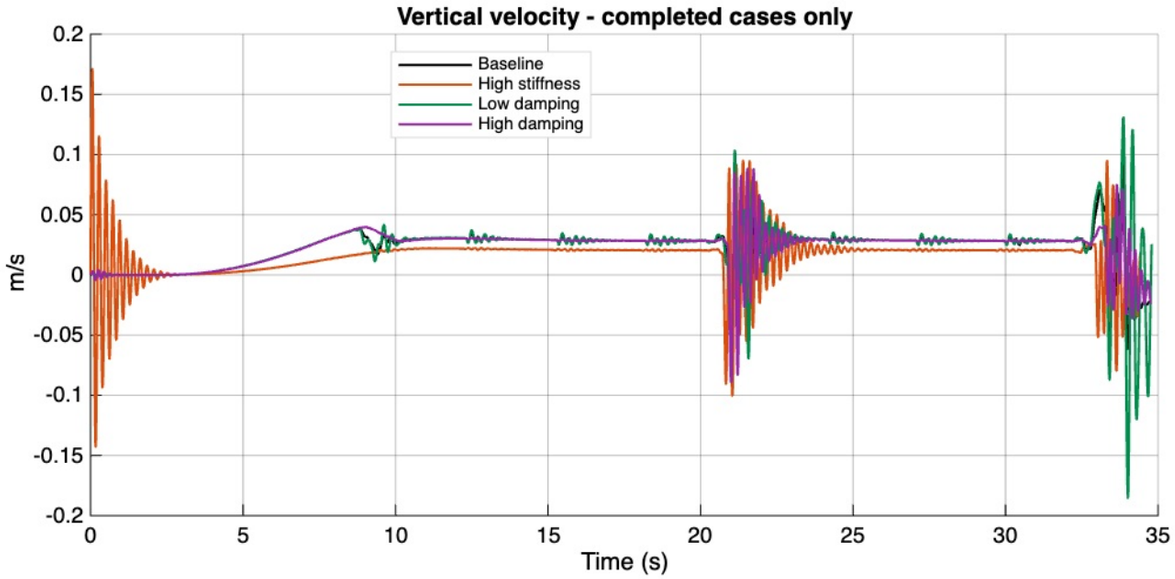

Pitch angle - completed cases only
deg
Time (s)
Baseline
High stiffness
Low damping
High damping

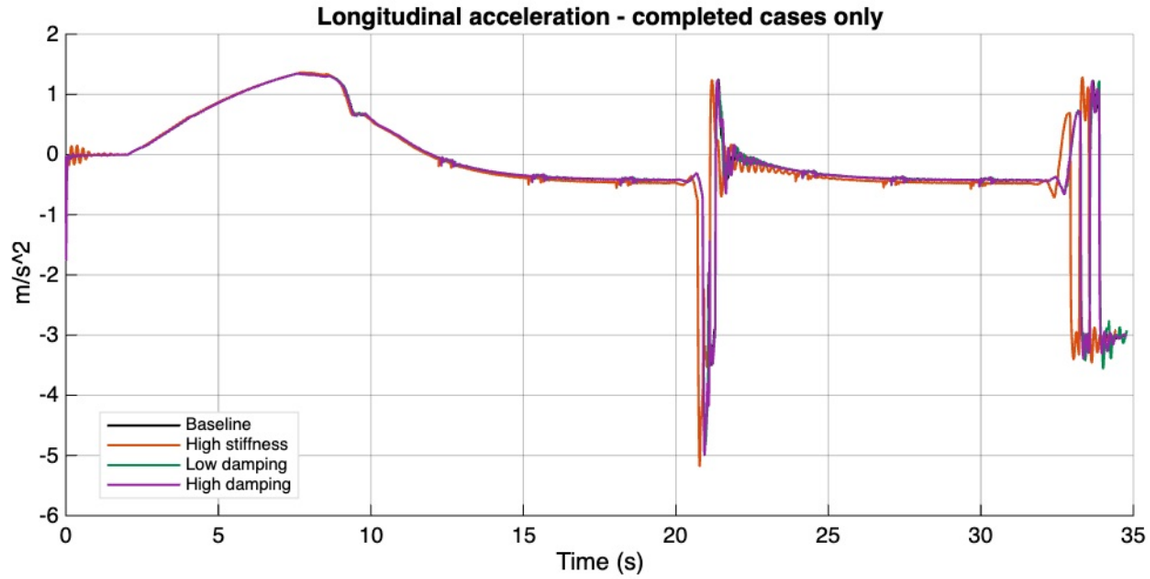

Roll rate - completed cases only
deg/s
Time (s)
Baseline
High stiffness
Low damping
High damping

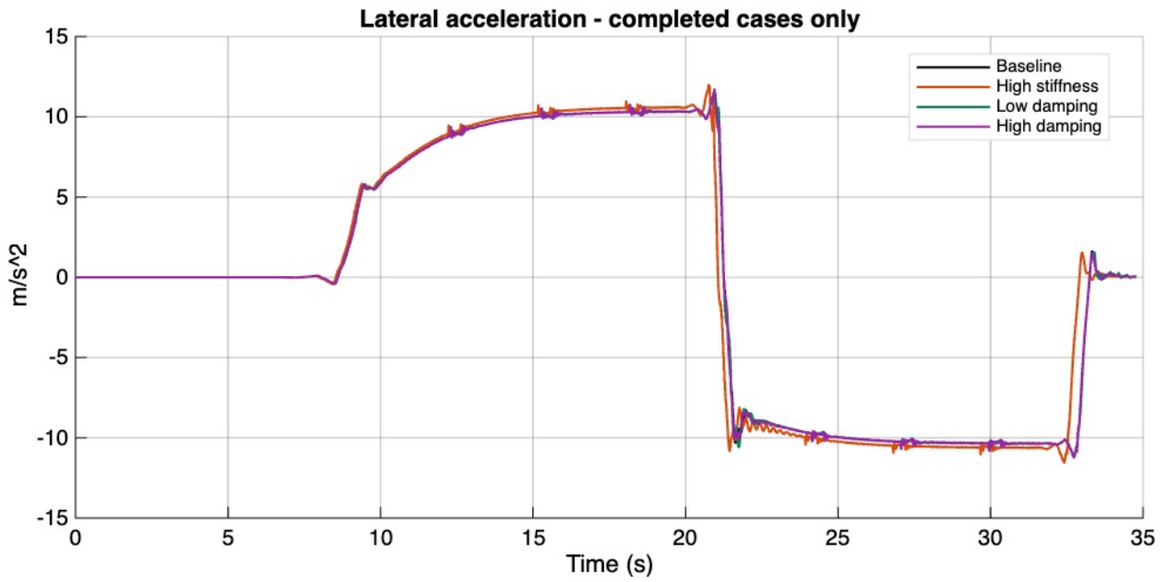

Pitch rate - completed cases only
deg/s
Time (s)
Baseline
High stiffness
Low damping
High damping

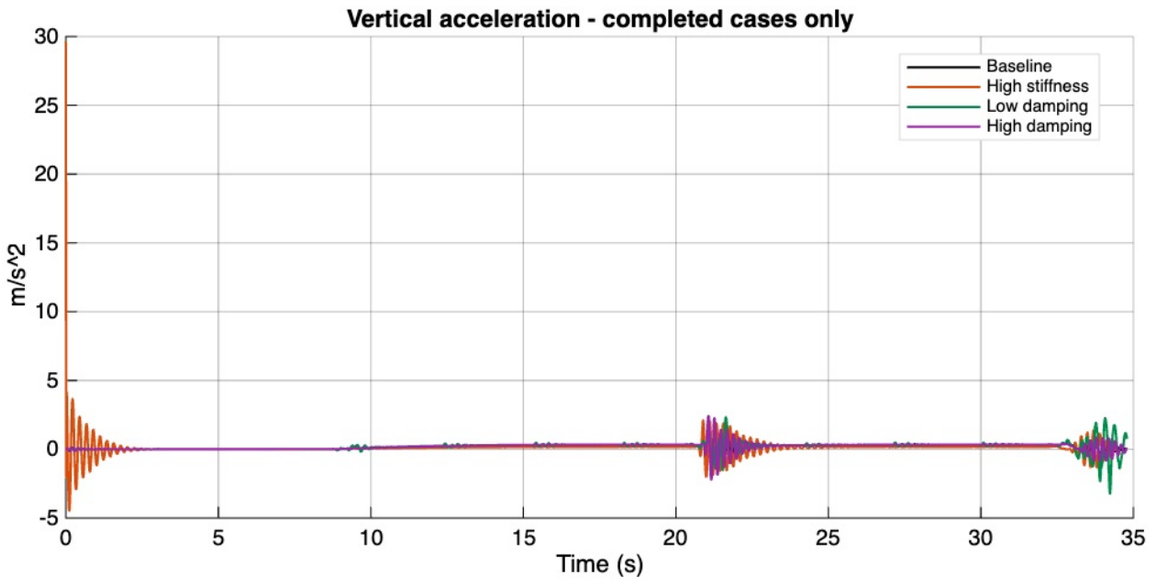

Yaw rate - completed cases only
deg/s
Time (s)
Baseline
High stiffness
Low damping
High damping

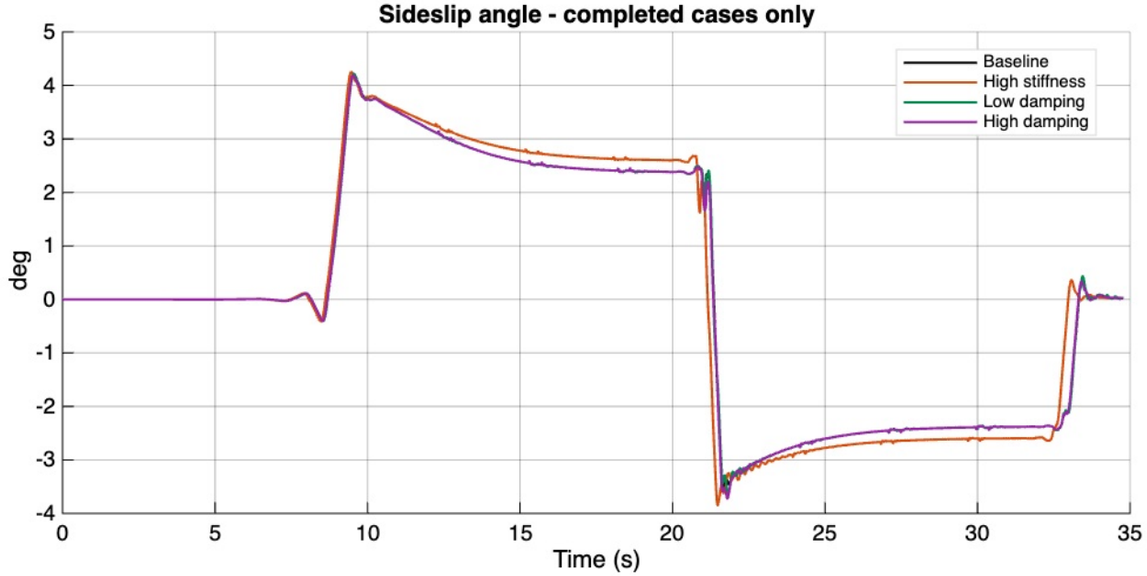

Longitudinal velocity - completed cases only
m/s
Time (s)
Baseline
High stiffness
Low damping
High damping

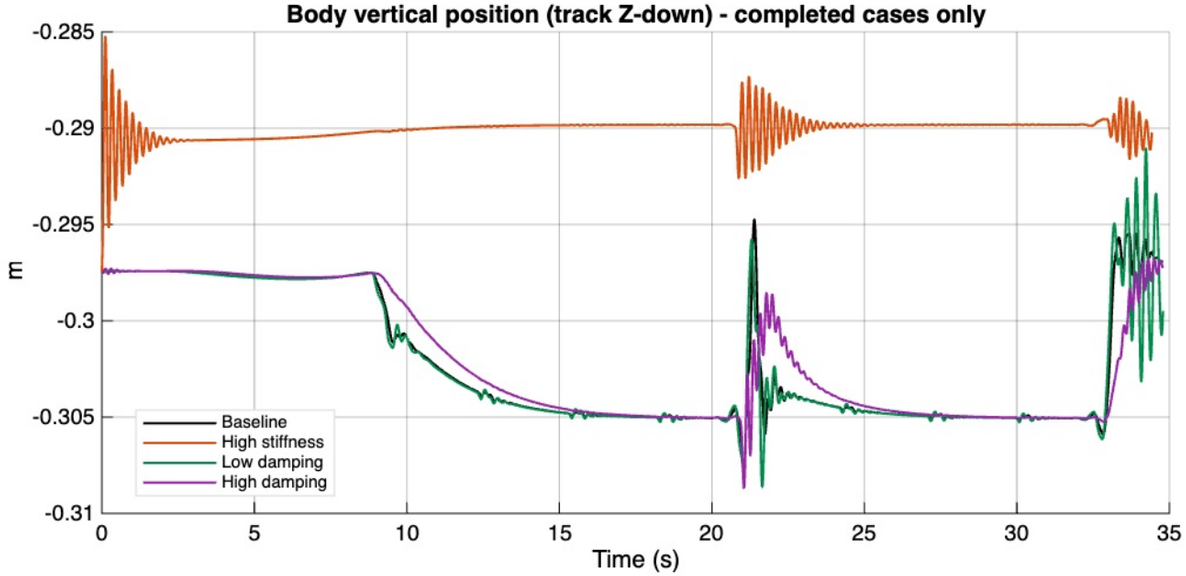

Vertical velocity - completed cases only
Baseline
High stiffness
Low damping
High damping
m/s
Time (s)

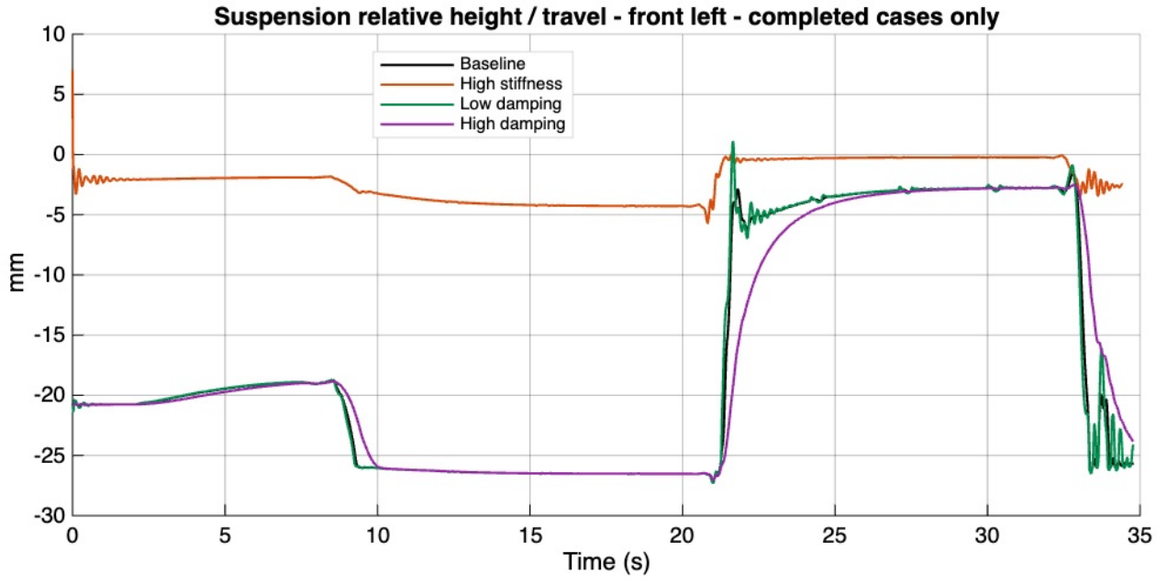

Longitudinal acceleration - completed cases only
Baseline
High stiffness
Low damping
High damping
m/s^2
Time (s)

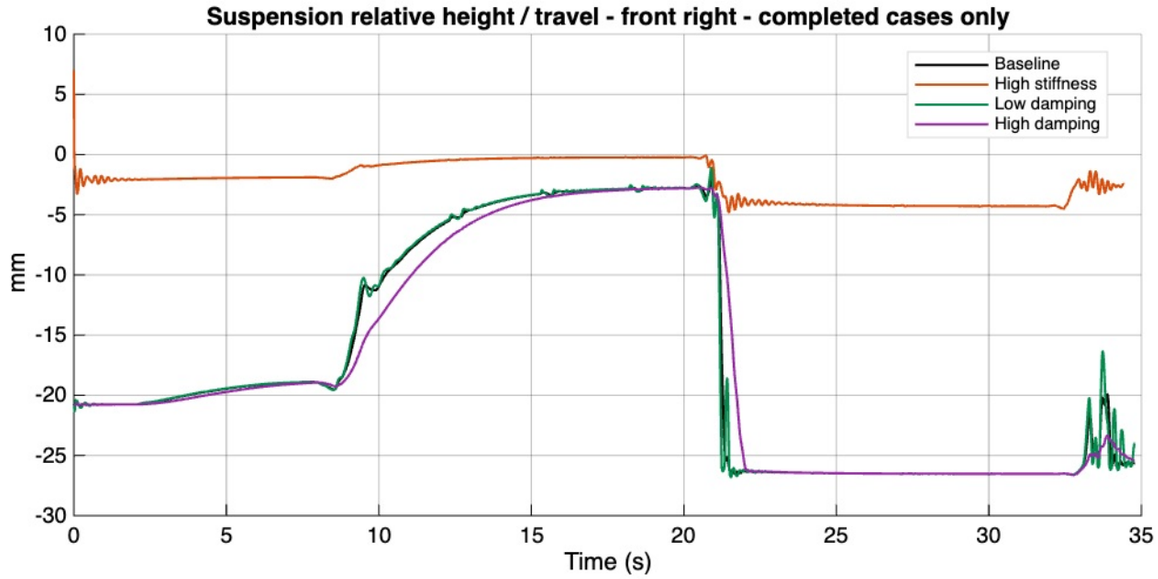

Lateral acceleration - completed cases only
Baseline
High stiffness
Low damping
High damping
m/s^2
Time (s)

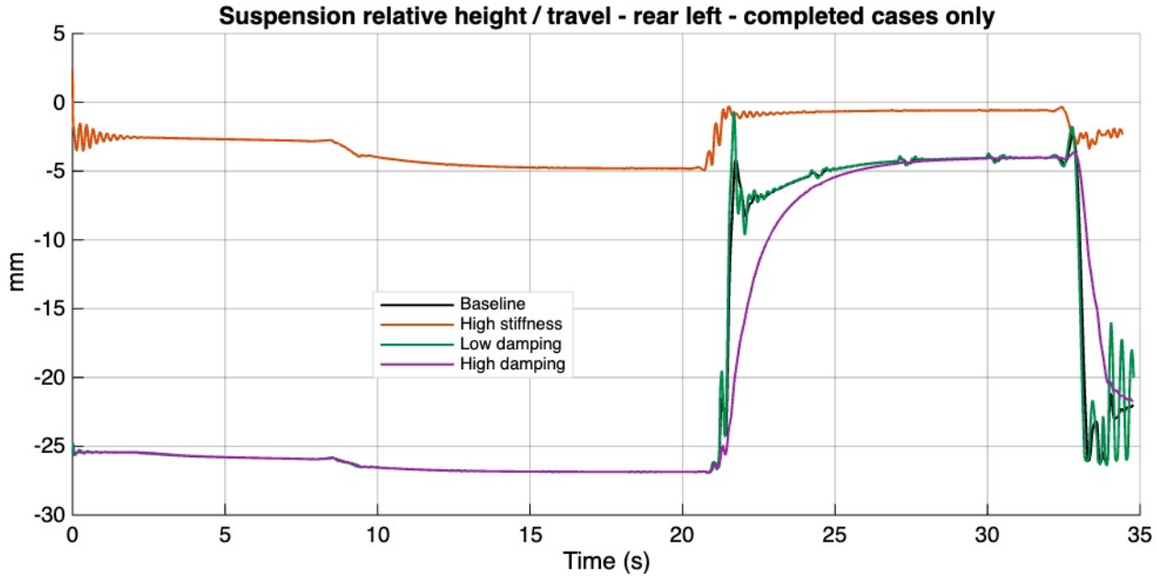

Vertical acceleration - completed cases only
Baseline
High stiffness
Low damping
High damping
m/s^2
Time (s)

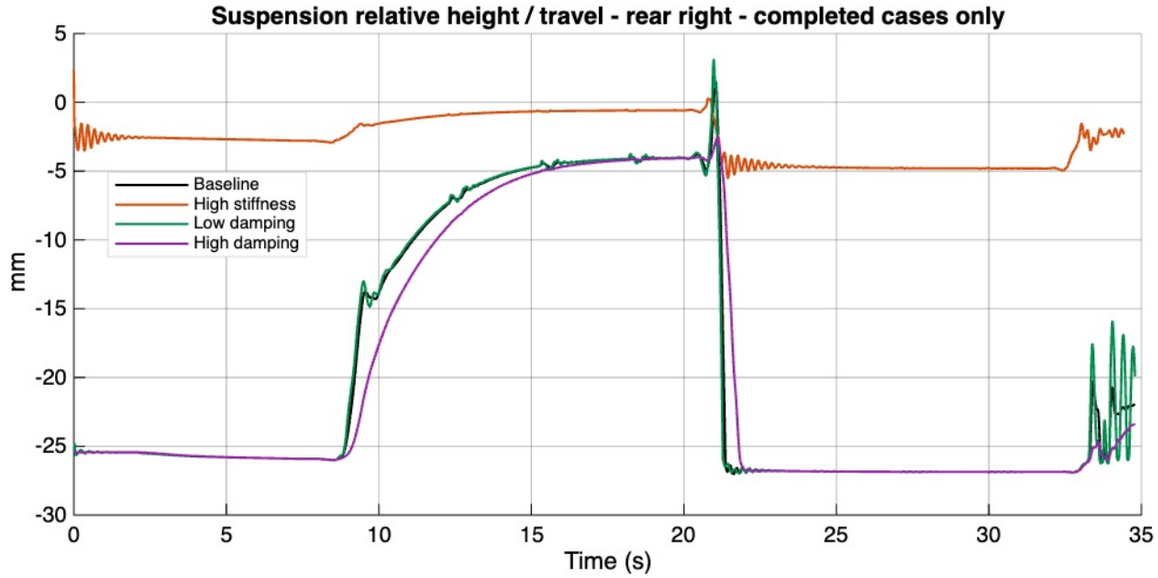

Sideslip angle - completed cases only
Baseline
High stiffness
Low damping
High damping
deg
Time (s)

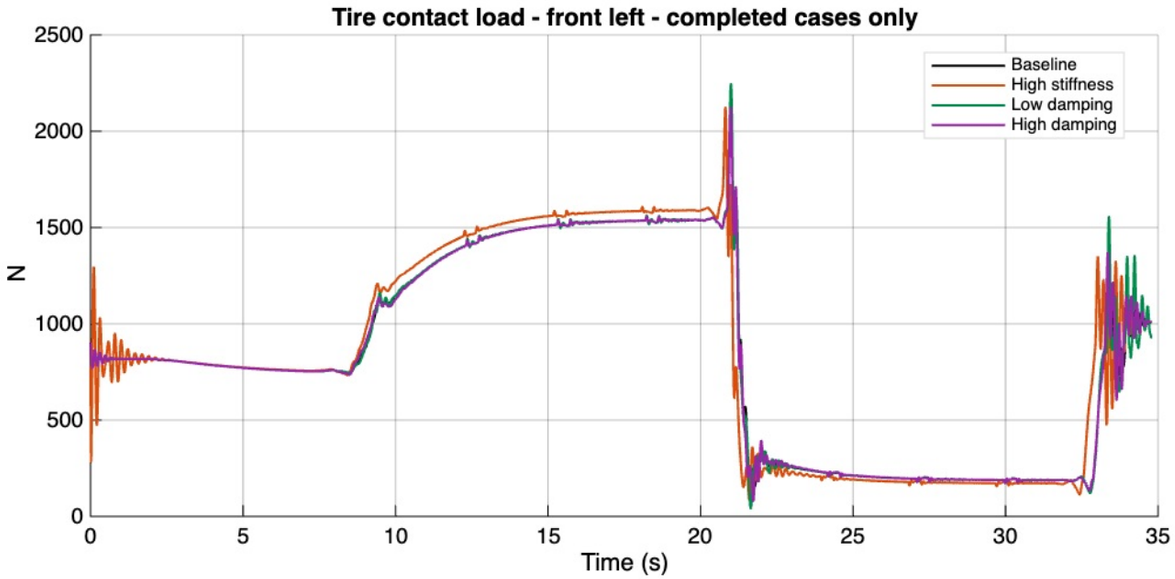

Body vertical position (track Z-down) - completed cases only
Baseline
High stiffness
Low damping
High damping
m
Time (s)

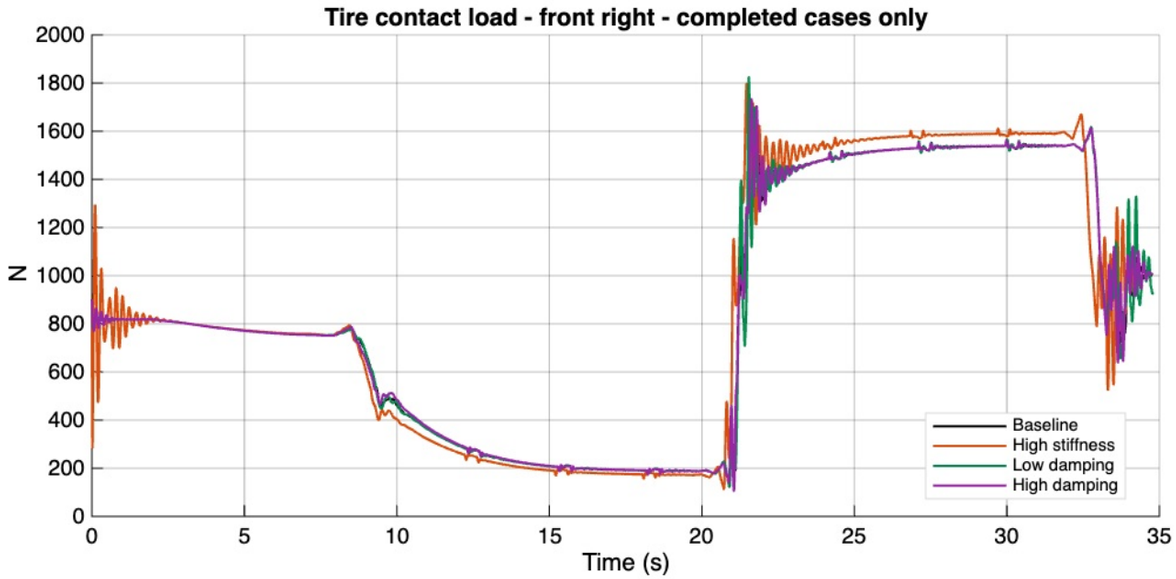

Suspension relative height / travel - front left - completed cases only
Baseline
High stiffness
Low damping
High damping
mm
Time (s)

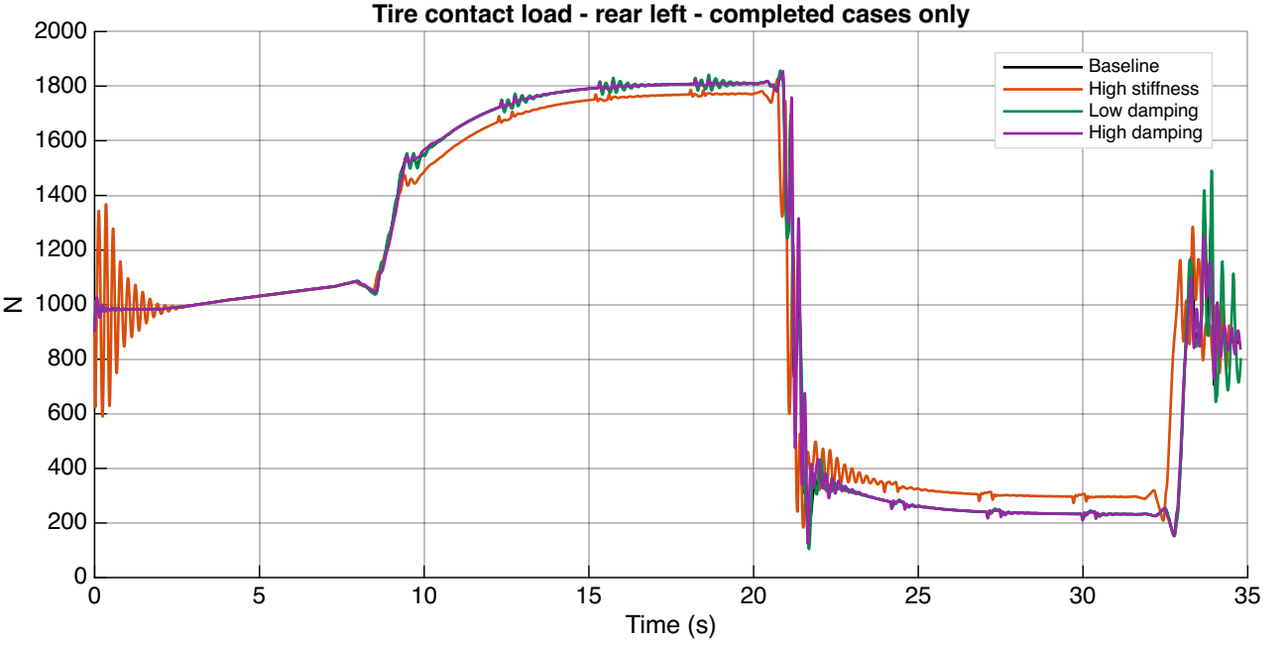

Suspension relative height / travel - front right - completed cases only
Baseline
High stiffness
Low damping
High damping
mm
Time (s)

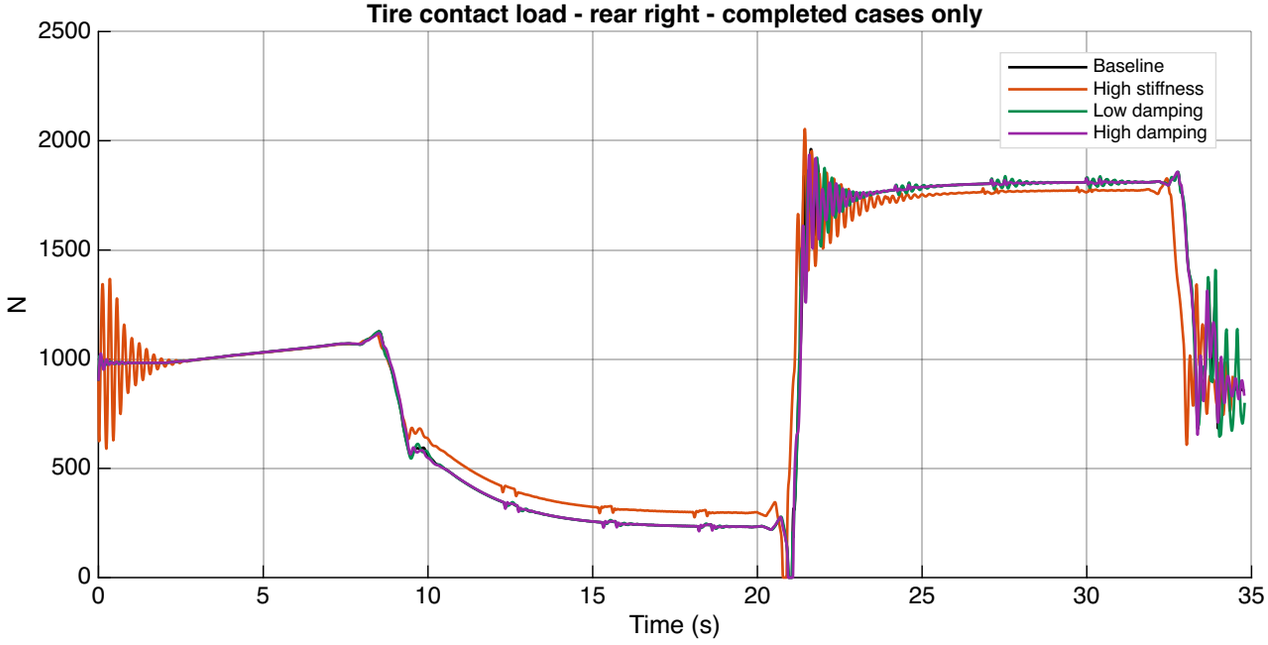

Suspension relative height / travel - rear left - completed cases only
Baseline
High stiffness
Low damping
High damping
mm
Time (s)

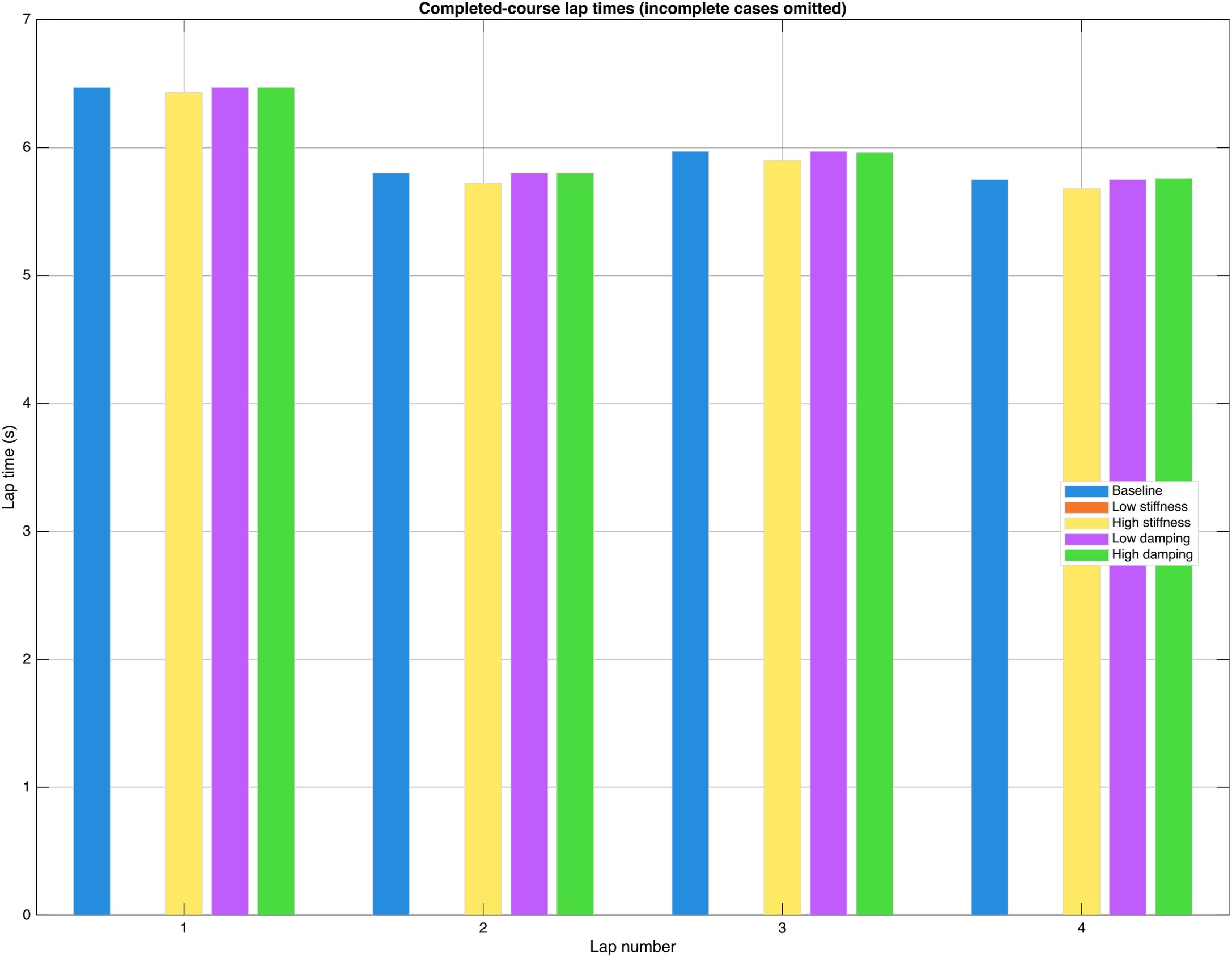

Suspension relative height / travel - rear right - completed cases only
Baseline
High stiffness
Low damping
High damping
mm
Time (s)


Tire contact load - front left - completed cases only
Baseline
High stiffness
Low damping
High damping
N
Time (s)


Tire contact load - front right - completed cases only
Baseline
High stiffness
Low damping
High damping
N
Time (s)


Tire contact load - rear left - completed cases only
Baseline
High stiffness
Low damping
High damping
N
Time (s)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30
35


Tire contact load - rear right - completed cases only
Baseline
High stiffness
Low damping
High damping
N
Time (s)
0
500
1000
1500
2000
2500
0
5
10
15
20
25
30
35


Completed-course lap times (incomplete cases omitted)
Lap time (s)
Lap number
0
1
2
3
4
5
6
7
1
2
3
4
Baseline
Low stiffness
High stiffness
Low damping
High damping